

Size- and Position-dependent opacity in Hungarian

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1 Introduction

Syntactic opacity or syntactic islandhood was originally proposed in an absolute sense (Ross 1967): a constituent is either transparent to all syntactic operations or opaque to all of them. However, opacity was later found to be occasionally *selective*: a constituent may be opaque to certain syntactic operations but transparent to others (e.g. Williams 1974, 2003, 2013; Keine 2016, 2019, 2020; Abels 2012; Wurmbrand 2001, 2015; Müller 2014a,b; Poole 2023; Meadows 2023).

Consider the Hungarian embedded clause containing the high sentential adverb *szerecsére* ‘fortunately’ in (1).¹ This embedded clause is not an absolute island, because relative pronouns can be subextracted from it.

- (1) *A város, [REL amelyik-en]₁ hall-ott-am, [hogy [TOP ma] [SADV szerecsére] keresztül fut-ott-ak __₁]*
the city which-SUP hear-PST-1SG.DEF that today fortunately across run-PST-3PL.INDF
‘The city which I heard that today they fortunately ran across.’

On the other hand, this embedded clause is not transparent to all movement types; it is selectively opaque. For instance, focus movement cannot proceed from within it:

- (2) *??[FOC (CSAK) A VÁROS-ON]₁ hall-ott-am, [hogy [TOP ma] [SADV szerecsére] keresztül fut-ott-ak __₁].*
only the city-SUP hear-PST-1SG.DEF that today fortunately across run-PST-3PL.INDF
Intended: ‘I heard that (*fortunately) today they ran across ONLY THE CITY.’

Previous work on selective opacity (e.g. Williams 2003; Abels 2012; Müller 2014a,b; Keine 2016, 2019, 2020; Poole 2023; Meadows 2023) restricted its attention to the *internal structural size* of constituents while determining their opacity. For example, the Williams Cycle in (3) (Williams 1974, 2003) is a generalization on *size-dependent opacity*: Structurally larger clauses are opaque to more syntactic operations than smaller ones, where β P clause is larger than a γ P clause iff β is higher than γ in some functional sequence.

- (3) **Williams Cycle/Size-dependent opacity:** Movement to Spec, β P cannot proceed across a dominating α P, where $\alpha > \beta$ in the functional sequence. (Williams 2003; Poole 2023)

The Hungarian sentence in (2) exhibits *size-dependent* opacity in (3) if has the structure in (4): (i) *hogy* ‘that’ is an adjunct to the clause (i.e. not a head in *fseq*) and thus does not affect clause size in the relevant sense; (ii) focus movement lands in matrix Spec, γ P and $\beta > \gamma$ in *fseq*; (iii) the addition of the adverb changes the size of the embedded clause from γ P to β P. (These assumptions will be supported later.) At the same time, (1) has the structure in (5): Since the relative pronoun movement lands in a position preceding the adverb in a Spec, α P such that $\alpha > \beta$, sentential adverbs do not block the relative pronoun movement.

- (4) $[_{\gamma P} \times \dots [_{VP} \dots [_{\beta P} \text{ sentential adverbs } [_{\gamma P} \dots \text{XP} \dots]]]]$ (5) $[_{\alpha P} \checkmark \dots [_{VP} \dots [_{\beta P} \text{ sentential adverbs } [_{\gamma P} \dots \text{XP} \dots]]]]$

¹In the bracket diagrams occurring above Hungarian examples, embedded clauses are highlighted in gray, dashed arrows refer to subextraction from embedded clauses, and simple arrows refer to all other types of movement. If β P is a projection in a matrix clause, β P' is a projection in an embedded clause, and β P'' is a projection in a doubly-embedded clause. In first line of the glossing, the various syntactic positions (focus, particle, relative etc.) are marked with subscripts. The glossing follows the Leipzig Glossing rules. The abbreviations are the following: 1 = first person, 2 = second person, 3 = third person, ACC = accusative, SADV = sentential adverb, COND = conditional, DEF = definite, FOC = focus, IN = inessive, INDF = indefinite, INF = infinitive, INS = instrumental, PART = particle, PL = plural, POT = potential, PST = past, Q = quantifier, REL = relative, SG = singular, SUB = subrelative, SUBJ = subjunctive, SUP = superessive, TOP = topic

The uncited Hungarian data comes from my own data collection. Although I am a native speaker of Hungarian, the grammaticality of critical sentences were double-checked in two surveys. The Hungarian data of the paper comes from the standard dialect spoken in Budapest. The first one was conducted in Summer, 2022, the second one in Spring, 2023. The surveys had 15 and 35 native-speaker participants, respectively.

However, *internal size* is not the only factor determining the selective opacity of clauses in Hungarian. The matrix *position* in which the embedded clause ends up also constrains the set of movement types that are grammatical from within this clause. In particular, fewer movement types can proceed from embedded clauses that themselves end up higher in the matrix clause. For instance, while the default position of Hungarian embedded clauses is post-verbal as in (2), embedded clauses can move to a topic position preceding sentential adverbs, i.e. c-commanding βP as in (6). However, foci may not originate in a clause that later gets topicalized to a higher position, even if it would be grammatical without the leftward movement of the clause. Put differently, after focus movement has happened, the embedded clause cannot undergo remnant movement.

- (6)
$$[\dots [\text{hogy} \dots \text{XP}]]_{\beta P} \dots [\text{VP} \times [\dots [\text{VP} \text{ V } [\text{hogy} \dots \text{XP}]]]]]$$
- $\downarrow$ ----- $\uparrow$
 $[_{TOP} \text{Hogy}]_{\langle FOC \rangle} \text{ CSAK } A \text{ VÁROS-ON} \rangle_2 \text{ fut-ott-am} \text{ keresztül } _2]_1 [_{SADV} \text{ szerencsére}] \text{ mindenki } \langle FOC \rangle^* \text{ CSAK}$
 that only the city-SUP run-PST-1SG.INDF across fortunately everyone only
 $A \text{ VÁROS-ON} \rangle_2 \text{ hall-ott-a} \text{ } _1$
 the city-SUP hear-PST-3SG.DEF
 ‘That I ran across ONLY THE CITY, fortunately everyone heard it.’

Although earlier work has handled moved constituents as islands (e.g. Ross 1967; Wexler and Culicover 1983), more recent works have argued that subextraction from moved constituents may be possible (Sauerland 1999; Collins 2005). Thus, βP -sister topic clauses are not absolute islands in Hungarian. Relative pronouns can be extracted from topicalized clauses as in (7) just as it was possible from postverbal clauses like (1).

- (7)
$$[_{\alpha P} \checkmark [\dots [\text{hogy} \dots \text{XP}]]_{\beta P} \dots [\text{VP} \text{ V } [\text{hogy} \dots \text{XP}]]]]$$
- $\downarrow$ ----- $\uparrow$
 $A \text{ város, } [_{REL} \text{ amelyik-en}]_2 [_{FOC} \text{ hogy keresztül fut-ott-am } _2]_1 [_{SADV} \text{ szerencsére}] \text{ mindenki}$
 the city which-SUP that across run-PST-1SG.INDF fortunately everyone
 $\text{ hall-ott-a } _1.$
 hear-PST-3SG.DEF
 ‘The city, which, that we ran across it, everyone fortunately heard it.’

The generalization on *position-dependent opacity* that I will argue for is the following:

- (8) **Position-dependent opacity (simplified):** Movement of the clause cannot be remnant movement.

Existing theories of selective opacity (e.g. Williams 2003; Abels 2012; Müller 2014a,b; Keine 2016, 2019, 2020; Poole 2023; Meadows 2023) focus on constituent-internal *structural size*, and hence cannot fully capture the facts of *position-dependent opacity* in (8). Therefore, my theoretical account relies on the idea that embedded clauses need to get into a certain matrix *position* in order to become transparent; given bottom-up structure building, this restricts possible landing sites of movement in a manner reminiscent of the Williams Cycle in (3).

The key assumptions of my analysis are the following: Firstly, building on the intuition originally proposed by Chomsky (1977: 85) and generalized by Bošković (2014), I propose that movement steps must be internal to (clausal) extended projections. Therefore, no movement step can start within the embedded clause and land in the matrix clause. Secondly, although successive cyclic movement via clause edges is a strategy to keep movement internal to extended projections (intuitively, the edge does not belong to the embedded extended projection), I argue that Hungarian clauses do not have edge positions.

The alternative strategy that Hungarian uses to keep movement internal to extended projections is *the re-merger of embedded clauses with an equal-sized matrix projection*. In (9), the embedded $\gamma P'$ re-merges with the matrix γP in a way such that the matrix projections dominating the re-merger site count as a continuation to the embedded extended projection. This is illustrated in (9): movement from $\gamma P'$ to the matrix Spec, βP can proceed as βP dominates the re-merger site of $\gamma P'$. In (10), a larger $\alpha P'$ re-merges with the matrix αP .

- (9)
$$[_{\beta P} \checkmark [_{\gamma P} [_{\gamma P'} \dots \text{XP}]]_{\gamma P} \dots [\text{VP} \text{ V } [_{\gamma P'} \dots \text{XP}]]]]$$
- $\downarrow$ ----- $\uparrow$
 $[_{\alpha P} [_{\alpha P'} \dots \text{XP}]]_{\alpha P} \dots [\text{VP} \times [\text{VP} \text{ V } [_{\alpha P'} \dots \text{XP}]]]]]$

Size- and position-dependent opacity follow from properties of the re-merger strategy. On the one hand, unlike "unlocking operations" in other accounts (e.g. Rackowski and Richards 2005; Van Urk and Richards 2015; den Dikken 2018), the re-merger operation is size-sensitive. Thus, an embedded $\gamma P'$ re-merges with the matrix γP in (9), but an embedded $\alpha P'$ re-merges with the matrix αP in (10). If $\alpha P > \beta P > \gamma P$ in the extended projection and transparency is gained only *after* re-merger, the $\alpha P'$ becomes transparent to movement to matrix positions *above* Spec, β , i.e. $\alpha P'$ is opaque to movement to Spec, βP . In other words, *size-dependent opacity* in (3) is derived because larger embedded clauses can re-merge and become transparent later in the derivation. In (2), since clauses with sentential adverbs re-merge above the landing site of focus movement, focus movement cannot proceed from such clauses. In contrast, relative movement in (1) lands above the re-merger site of clauses with sentential adverbs.

On the other hand, since the embedded clause and the matrix clause projection with which the embedded clause re-merges are of the same size and category, their merger is symmetric and they project together. Under the assumption that non-maximal projections cannot move (e.g. Chomsky 1995), re-merged embedded clauses are frozen in the re-merger site. In other words, *position-dependent opacity* in (8) is derived because embedded clauses are immobile after they become transparent to movement to matrix positions via re-merger. In (6), if focus movement can originate in the embedded clause, the clause must re-merge below the landing site of focus movement. However, if such re-merger happens, the clause cannot be topicalized to a position preceding the focus. At the same time, since topicalized clauses re-merge in a position below the landing site of relative movement, relative movement landing higher can proceed from topicalized clauses in (7).

Thus, compared to previous approaches to selective opacity, the first novel aspect of my proposal is that clauses are not only opaque because they are internally large but also because they are only mobile prior to their re-merger (and transparency) in the matrix clause. Another novel aspect of the analysis is that it unifies the various kinds of selective opacity and the more standard notions locality (subjacency, barriers, phases etc.). In particular, I predict that if the successive cyclic strategy is available in some domain, size- and position-dependent do not arise in that domain. As we will see, this prediction is borne out: Mongolian exhibits successive cyclicity in the clausal domain alongside the violations of (3) (Fong 2019). Furthermore, Hungarian exhibits successive cyclicity in the nominal domain alongside absence of the ban on movement of nominal remnants (i.e. there is no nominal parallel to (8)). The unification of these is not trivial, and previous accounts mostly handled these as independent notions (e.g. Keine 2019).

The discussion proceeds as follows: In Section 2, I demonstrate that the various movement types in Hungarian exhibit size-dependent opacity (Williams Cycle effects). Section 3 introduces novel data on the basis of which I claim that Hungarian also exhibits position-dependent opacity. In Section 4, demonstrates that position-dependent opacity cannot be reduced to size-dependent opacity, and thus accounts of size-dependent opacity do not straightforwardly derive position-dependent opacity. In Section 5, I propose that Hungarian embedded clauses become transparent via re-merger, which provides a unified account for size- and position-dependent opacity. Although my proposal does not automatically account for every instance of selective opacity, it opens new potential avenues for treating selective opacity and locality in a unified way.

2 Size-dependent opacity in Hungarian

Among other factors, the opacity of Hungarian clauses is determined by their structural size; larger clauses are opaque to more syntactic operations. Section 2.1 introduces the relevant left-periphery projections and cross-clausal movement types in Hungarian. Section 2.2 argues that topics, quantifiers, and the complementizer *hogy* 'that' *do not* have dedicated heads in the functional sequence of Hungarian clauses, that is, these constituents do not contribute to the structural size of clauses in the relevant sense. Section 2.3 introduces the first half of the data on the basis of which I claim that the Hungarian has size-dependent opacity. More specifically, the section describes which cross-clausal movement operations can cross which left-periphery positions. Section 2.4 extends the scope of the generalization with data on cross-clausal movement to the opacity differences between clause-types (subjunctive vs. tensed). The discussion is concluded by Section 2.5 that re-frames the generalization on size-dependent opacity as a stricter version of the cross-linguistically observed generalization, the Williams Cycle (Williams 1974, 2003, 2013).

2.1 Projections in Hungarian left-periphery

This subsection introduces the Hungarian clausal left-periphery summarized in (11): TP hosts verbal particles; Foc(us)2P and Foc(us)IP host foci; Com(mitment)P hosts sentential adverbs; and Rel(ative)P hosts relative pronouns. The projections in (11) will be discussed in a lower-to-higher order.

- (11) [_{RelP} relative pronouns [_{ComP} sentential adverbs [_{Foc2P} foci [_{Foc1P} foci [_{TP} particles[VP]]]]]]

The lowest relevant left-periphery position is the position of verbal modifiers including particles (e.g. Horvath 1981; Koopman and Szabolcsi 2000; É. Kiss 2006). For instance, the particle *keresztül* ‘across’ can move to this pre-verbal position in (12). The particle *keresztül* is base-generated as a postposition in a post-verbal and VP-internal PP, where it assigns superessive case to its complement. In Dékány and Hegedűs’ (2015) analysis, particle movement is treated as phrasal movement rather than head movement: After *keresztül*’s complement has moved via some Spec,*pP* edge position, the PP-remnant containing only *keresztül* can independently move out of the postpositional extended projection and land higher in the clause. Following Olsvay (2000), É. Kiss (2008), and Surányi (2009; 2012), the preverbal (but still relatively low) landing site for particles will be referred to as Spec,TP such that T>V in the *f*(unctional) *seq*(uence).²

- (12) [_{TP} ✓ [_{VP} [_{PP} ... XP ...]]]
- [_{PART} *Keresztül*]₁ *fut-ott-am* [_{PP} *a város-on* ____]
across run-PST-1SG.INDF the city-SUP
‘I ran across the city.’

Verb particles like *keresztül* ‘across’ are capable of moving cross-clausally and landing in the matrix Spec,TP, where they precede the matrix verb as in (13) (Farkas and Sadock 1989). The cross-clausal movement of particles is also known as *particle climbing*.

- (13) [_{TP} ✓ ... [... [_{VP} [_{PP} ... XP ...]]]]
- [_{PART} *Keresztül*]₁ *akar-t-am*, [*hogy fus-s-anak a város-on* ____]
across want-PST-1SG.INDF that run-SUBJ-3PL.INDF the city-SUP
‘I wanted them to run across the city.’

Foci and *wh*-constituents both have to move to a position preceding the particle. In the literature, these have been assumed to land in the same position, which has been generally referred to as Spec,FocP (e.g. Horvath 1981, 1986; É. Kiss 1987; Brody 1990). Focus movement is exemplified in (14). Whenever the focused constituent lands in Spec,FocP, the verb moves to the Foc head and therefore precedes the particle (Brody 1990).

- (14) [_{FocP} ✓ Foc+T+V[_{TP} T+V [_{VP} V ... XP ...]]]
- [_{FOC} (*CSAK*) *A VÁROS-ON*]₂ *fut-ott-am*₁ [_{PART} *keresztül*]₃ ____₁ ____₂ ____₃
only the city-SUP run-PST-1SG.INDF across
‘I have ran across ONLY THE CITY’

Hungarian *wh*-movement lands in the same Spec,FocP position and triggers the same verb-particle inversion (Horvath 1986):

- (15) [_{FocP} ✓ Foc+T+V[_{TP} T+V [_{VP} V ... *wh*-XP ...]]]
- [_{FOC} *Melyik város-on*]₂ *fut-ott-ál*₁ [_{PART} *keresztül*]₃ ____₁ ____₂ ____₃?
which city-SUP run-PST-2SG.INDF across
‘Which city did you run across?’

²Particle movement often influences the telicity or *Aktionsart* of predicates (É. Kiss 2006; Hegedűs 2020). The exact motivation and landing site of particle movement is an open question in the Hungarian literature. The position of the particle has been identified as Spec,AspP (e.g. É. Kiss 2002) Spec,PredP (e.g. Csirmaz 2004), or simply Spec,PartP (Egressy to appear). Although pre-verbal modifiers have been claimed to influence telicity and aspect as well as other semantic properties (É. Kiss 2006), it is not clear whether there is a unified way to describe the properties of all preverbal particles (Komlósy 1992). According to Komlósy (1992) and Szendrői (2003), at least for a subset of verbs, the movement happens if the verb cannot bear phonetically neutral sentential stress.

Focus-movement and *wh*-movement can cross clause boundaries and land in the matrix Spec,FocP as in (16). The sentence in (16) also shows that in the embedded clause, the particle occurs pre-verbally in Spec,TP, which is expected if the long-moving focus does not stop in the embedded Spec,FocP.³

- (16) $\checkmark$ -----₁
 [FocP $\checkmark$... [... [VP [PP ... *wh*-XP ...]]]]
 [Foc Melyik város-on]₂ akar-t-ad₁ _____₁, [hogy [PART keresztül]₃ fus-s-anak _____₂ _____₃]?
 which city-SUP want-PST-2SG.INDF that across run-SUBJ-3PL.INDF
 ‘Which city did you want them to run across?’

In this paper, focus and *wh*-movement are treated as instances of the same operation: Whatever is stated for one also applies to the other.

Hungarian clauses can have multiple foci as in (17) (É. Kiss 1998:p.16). É.Kiss (1998) proposes that such constructions have two focus projections, where FocIP is dominated by Foc2P, headed by Foc1 and Foc2, respectively. In (17), the verb undergoes head-movement first from T to Foc1 then from Foc1 to Foc2 in a way that it ends up between the two foci.

- (17) $\checkmark$ -----₁
 [Foc2P XP [Foc2+ $\checkmark$ [Foc1P YP [Foc1+T+V [TP particle [TP T+V ... XP ... YP]]]]]]]
 [Foc2 CSAK KÉT LÁNY]₁ olvas-ott-Ø₂ [Foc1 CSAK EGY KÖNYV-ET]₃ _____₂ [PART el] _____₂ _____₁ _____₃
 only two girl read-PST-3SG.INDF only one book-ACC away
 a vizsgá-ra
 the exam-SUB
 ‘ONLY TWO GIRLS were such that they read ONLY ONE BOOK for the exam.’

The distinction between the two focus positions is only possible in clauses with at least two foci. Therefore, in structures with a single focus, I will not make a distinction between landing in Spec,Foc1P and Spec,Foc2P, but use the neutral term Spec,FocP.⁴

The next highest position is the position of sentential adverbs like *szerencsére* ‘fortunately’, *sajnos* ‘unfortunately’, and *következésképpen* ‘consequently’, which always precede foci in the left-periphery as in (18) (Horvath 1981; É. Kiss 1987; Egedi 2021). These adverbs do not modify the truth conditional content of the proposition but specify the speaker’s attitude towards the proposition and locate the sentences in the discourse. If these high adverbs occur in embedded clauses, the embedded clause must describe a speech event, a commitment to a proposition or the evaluation of a proposition rather than just a proposition. In other words, embedded clauses containing high sentential adverbs must describe the content of some verbal message. For instance, if *szerencsére* ‘fortunately’ occurs in the complement clause of *hall* ‘hear’ in (18), the subject of the matrix clause must have heard someone’s positive report (fortunate news) of an event rather than the event itself. Krifka (2023) introduces three projections JudgeP, Com(mitment)P, and ActP to host high sentential adverbs. For the sake of simplicity, I assume that all high sentential adverbs are hosted in Spec, ComP although they could be analyzed as ComP-adjuncts as well. The importance of these adverbs is that they indicate that the clause with the adverb has a ComP layer. The ordering of these adverbs with respect to each other will not be

³Here I focus on genuine cross-clausal movement. Since Hungarian allows the *pro*-drop of pronouns bearing nominative and accusative, whenever some operator is co-indexed with a nominative- or accusative-marked trace, it is possible that the operator is base-generated in the matrix clause, while the empty category in the embedded clause is a *pro*-dropped resumptive pronoun bound by the operator in the matrix clause (for further discussion, see Jánosi 2013; Den Dikken 2018). For this reason, I avoid the discussion of any cross-clausal movement leaving nominative- or accusative-bearing traces. Thus, I don’t discuss subject- and object-raising. Instead, this paper mostly uses the extraction of the superessive arguments like *a város-on* ‘the city-SUP’, *melyik város-on* ‘which city-SUP’, *amelyik-en* ‘which-SUP’ from the complement position of *keresztül* ‘across’. The head *keresztül* ‘across’ idiosyncratically assigns the superessive case to its complement. In isolation, the superessive case combined with ‘the city’ has the near-nonsense meaning ‘on the city’. Furthermore, the superessive suffix cannot be dropped even if its pronoun undergoes *pro*-drop (Marác 1989). Because of these, whenever we get the ‘across the city’ interpretation, we have a good reason to believe that *a város-on* is neither a matrix clause adjunct nor a proleptic argument but originates in the VP-internal PP headed by *keresztül* ‘across’ (thanks to Katalin É. Kiss and Marcel den Dikken for the helpful discussion). Moreover, I do not discuss cases of *wh*-scope marking/partial *wh*-movement, which is usually preferred by speakers over cross-clausal *wh*-movement. According to Horváth’s (1997) analysis of scope-marking constructions, scope marking and genuine cross-clausal movement involve different mechanisms.

⁴In this paper, clauses with more than two foci are not discussed. Foc1P and Foc2P could host more than two foci if only the highest focus occurs in Spec,Foc2P while all the other foci occur in Spec,Foc1P. At the same time, I do not know of any principled ground to exclude further projections like Foc3P, Foc4P etc.

central to this paper. Krifka (2023) locates ComP above layers modifying the propositional content of the clause, thus, it is reasonable to locate ComP above FocP as foci influence the truth conditions of clauses.⁵

- (18) $[_{FOC} MA]$ *hall-ott-am*, $[\text{hogy } (_{SADV} \text{ szerencsére}) [_{FOC} (CSAK) A \text{ VÁROS-ON}]_2 \text{ fut-ott-ak}_1$
 today hear-PST-1SG.DEF that fortunately only the city-SUP run-PST-3PL.INDF
 $[_{PART} \text{ keresztül}]_3 \text{ } _1 _2 _3]$.
 across
 Without *szerencsére*: ‘I heard that they ran across ONLY THE CITY.’ (i.e. the noise of running)
 With *szerencsére*: ‘I heard that someone said that fortunately they ran across ONLY THE CITY.’

As further addressed in Section 2.3, another environment in which ComP-sized clauses occur is the complement position of manner of speech verbs, where the person who makes the matrix utterance distances themselves from the embedded proposition.

Similarly to languages such as Italian (Rizzi 1997) and Wolof (Torrence 2013), Hungarian relative pronouns do not land in the same position as *wh*-phrases and foci (Horvath 1981). As demonstrated in (19), while foci follow high sentential adverbs, relative pronouns precede them. Furthermore, unlike the movement of foci, the movement of relative pronouns is not accompanied by head movement of the verb. For reasons of simplicity, the position of relative pronouns will be referred to as Spec,RelP (although Rizzi’s (1997) Spec,ForceP would be appropriate, as well).

- (19) $[_{RelP} \checkmark [_{ComP} [_{FocP} [_{TP} [_{VP} [_{PP} \dots XP \dots]]]]]]$
- Ez az a város, $\langle_{SADV} \text{ *következésképpen/ *szerencsére} \rangle$ $[_{REL} \text{ amelyik-en}]_1 \langle_{SADV} \text{ következésképpen/}$
 this that the city, consequently/ fortunately which-SUP consequently/
 $\text{ szerencsére} \rangle$ $[_{FOC} CSAK \text{ PETI-VEL}] \text{ fut-ott-}\emptyset_2$ $[_{PART} \text{ keresztül}]_3 \text{ } _2 _1 _3$
 fortunately only Peti-INS run-PST-3SG.INDF across
 ‘This is the city, which consequently/fortunately I ran across ONLY WITH PETI.’*

The relative pronoun movement can also cross clause boundaries and land in a matrix Spec,RelP:

- (20) $[_{RelP} \checkmark \dots [\dots [_{VP} [_{PP} \dots XP \dots]]]]$
- Ez az a város, $[_{REL} \text{ amelyik-en}]_1 \text{ az-t akar-t-ad, } [\text{hogy } [_{PART} \text{ keresztül}]_3 \text{ fus-s-anak } _1 _3]$
 this is the city which-SUP it-ACC want-PST-2SG.INDF that across run-SUBJ-3PL.INDF
 ‘This is the city which you wanted them to run across.’*

In sum, the clausal functional sequence at this point looks as follows:

- (21) Rel>Com>Foc2>Foc1>T>V

The reason for focusing on the heads in (21) and their projections is that the operators associated with them reliably represent structural height in the left-periphery e.g. sentential adverbs consistently precede foci, so it is reasonable to assume that a dedicated ComP dominates the dedicated FocPs. This is not trivial: As the following subsection argues, there are good reasons to believe that topics, quantifiers, and *hogy* ‘that’ can attach at various structural heights.

2.2 Topics, quantifiers, and *hogy* ‘that’ dedicated *fseq*-heads

Before turning to the size-dependent constraints on cross-clausal movement in Section 2.3, I briefly argue that there are no dedicated projections for topics, quantifiers, and complementizers i.e. (21) lacks the Top(ic)/Ref(erential), Q(uantifier)/Dist(ributive), and C heads.

The main motivation for ignoring topics and quantifiers is that only constituents with invariant structural height indicate the structural size of clauses reliably. In Hungarian, topics and positive quantifiers can occur before or after high sentential adverbs in neutral sentences, or even post-verbally (É. Kiss 2002; Egedi 2021):

⁵Further cross-linguistic support comes from Cinque’s hierarchy (Cinque 1999), in which evaluative adverbs are hosted in the Mood-Evaluative projection, which is one of the highest projections of the clause.

- (22) $\langle$ [_{TOP} *A diák-ok-at*]/ [_Q *sok diák-ot*] $\rangle$ [_{SADV} *sajnos*] $\langle$ [_{TOP} *a diák-ok-at*]/
the student-PL-ACC many student-ACC unfortunately the student-PL-ACC
[_Q *sok diákot*] $\rangle$ *el kell utasít-an-unk* $\langle$ [_{TOP} *a diák-ok-at*]/ [_Q *sok diákot*] $\rangle$
many student-ACC away need reject-INF-1PL the student-PL-ACC many student-ACC
'{The students/Many students}, we have to reject unfortunately.'

Furthermore, although quantifiers and topics are similar to foci in that they can occur in in finite clauses like (23), there is an essential difference: unlike foci, quantifiers and topics can also occur in radically truncated clauses lacking external arguments, verb inflection, and accusative case marking as (24) (Halm 2021). With Halm (2021), I analyze radically truncated clauses as bare VPs.

- (23) { [_{TOP} A kocsmá-ban]/ [_Q mindenhol]/ [_{FOC} CSAK A KOCSMÁ-BAN] } hangosít-juk fel a TV-t
the pub-IN everywhere only the pub-IN add.volume-1PL up the TV-ACC.
‘{ In the pub/ everywhere/ ONLY IN THE PUB}, we turn up the volume on the TV.’
- (24) [{ topics/quantifiers/*foci} [_{VP} [...]]]

- {[_{TOP} *Kocsmá-ban*]/[_Q *mindenhol*]/[_{FOC} **CSAK KOCSMÁ-BAN*]} *TV fel hangosít*
 pub-IN everywhere only pub-IN TV up add.volume
 ‘{ In the pub/ everywhere/ *ONLY IN THE PUB}, (someone) turns up the volume on the TV.’

On the basis of these, I follow Surányi (2002), Szendrői (2003), and É. Kiss (2010) in claiming that while particles, foci, sentential adverbs, and relative pronouns have dedicated projections in the left-periphery, quantifiers and topics do not occur in the specifiers of dedicated projections (i.e. there is no TopP and QP) but can adjoin to/merge with various maximal projections.⁶ Throughout the paper, the adjunction site of topics and quantifiers will be assumed to be maximal ComP and FocP projections in finite clauses like (23) (i.e. I set aside cases like (24)). It is a robustly established fact in the Hungarian literature (e.g. É. Kiss 1987, 1998, 2002; Szendrői 2003; É. Kiss 2010) that in the left-periphery no topics and positive quantifiers are c-commanded foci. Here, focus-topic-focus and focus-quantifier-focus orders are excluded by the assumption that the lowest adjunction site for topics and quantifiers is the outermost FocP. Topics and quantifiers will be discussed in greater detail in Section 4. (For a more extended discussion of topics and quantifiers, see Egressy to appear.)

Besides rejecting dedicated TopPs and QPs, I assume that the Hungarian clause structure does not have a dedicated CP projection for the complementizer. I follow Manetta's (2006; 2011) proposal for the Hindi-Urdu *ki* 'that', and claim that *hogy* 'that' is not a complementizer head in the clausal *fseq* but rather an edge marker, possibly inserted after spell-out. Instead of having a dedicated head in the clausal *fseq*, *hogy* 'that' will be handled as an adjunct to the maximal projection of clauses. For example, if an embedded clause has a sentential adverb but no relative pronoun, it is a bare ComP as in (25), whereas if it has a particle but no other left-periphery operator, it is a bare TP as in (26).⁷ This also means that the sister of the matrix verbs in (25) and (26) is not a CP but a ComP and a TP, respectively.

- (25) [_{VP} V [_{ComP} *hogy* [_{sentential adverb} [_{Com} [_{Foc2P} ...]]]]]
- (26) [_{VP} V [_{TP} *hogy* [_{particle} [_T [_{sMod} ...]]]]]

There are independent, Hungarian-specific arguments for handling *hogy* ‘that’ as an edge marker rather than a C(omplementizer) head selecting for a certain category of complement clauses. For example, Kenesei and Szeteli (2022 p. 98) bring a number of naturally occurring examples like (27) in which *hogy* precedes infinitival clauses. (However, this is an emerging tendency and not all speakers find these examples grammatical.)

⁶To be more precise, since negative existential quantifiers behave in the same way as foci (É. Kiss 1998), only quantifiers not belonging to this group will be assumed to be constituents lacking dedicated projections. In the case of topics, all subtypes will be assumed to be adjuncts.

⁷An alternative would be to adopt the idea in Radford (2018), according to which *that* occurring at the edge of English clauses of various sizes is always the realization of the highest head of the clause, possibly in a way that *hogy* keeps moving as long as the embedded left-periphery projects.

- (27) *Egy élet-et le-het-ne ar-ra áldoz-ni, [hogy az-ok-at meg ír-ni]*
 a life-ACC be-POT-COND it-SUB sacrifice-INF that that-PL-ACC PART write-INF
 ‘One could sacrifice a life for writing them.’

Furthermore, *hogy* ‘that’ can introduce either finite or subjunctive clauses (Section 2.4), and even embedded questions (Koopman and Szabolcsi 2000). In short, *hogy* does not seem to select for a certain clause-type or a certain syntactic category.

Secondly, *hogy* ‘that’ often occurs in front of constituents that are clearly non-propositional. Example (28) indicates that reaction words like *hoppá* ‘wow’ and *brrr* ‘brrr’ as well as greetings like *helló* and *szia* ‘hello’. Although the syntactic category of these words remains an open question, it is clear that they do not involve predication of any kind. Notice that this is different from the behaviour of English *that*, which is not compatible with greetings, e.g. **John did not say that hi.*

- (28) *Győző az-t (se) mond-t-a, [hogy brrr/ hoppá/ helló/ szia]*
 Győző it-ACC neither say-PST-3SG.DEF that brrr wow hello hello
 ‘Győző did (not even) say hi/ goodbye/ wow /brrr.’

Thirdly, *hogy* ‘that’ does not occur exclusively in complement clauses: In (29), *hogy* occurs in an adjunct clause preceding the conditional complementizer *ha* ‘if’.

- (29) *Örül-ök, [hogy ha jön-ssz]*
 be.happy-1SG.INDF that if come-2SG.INDF
 ‘I am happy if you come.’

Finally, the clause in which *hogy* occurs does not have to be a subordinate clause. Thus, *hogy* ‘that’ often occurs at the edge of *echo questions* like (30) (Anikó Lipták, p. c.).

- (30) A: *Mari jön-Ø este.* B: *Hogy ki jön este?!*
 Mari come-3SG.INDF evening that who come-3SG.INDF evening
 A: ‘Mari comes tonight.’ B: ‘Who comes tonight?!’

Thus, the morpheme *hogy* ‘that’ does not only occur at the edge of a single type of complement clauses, but shows a much wider distribution: The constituent following *hogy* does not have to be tensed, finite, or even propositional, and the constituents containing *hogy* do not have to be the complements of verbs. Non-projecting complementizers are not uncommon in the cross-linguistic literature, either: Embedded clausal spines base generated without a projecting complementizer have been independently proposed by Koopman and Szabolcsi (2000) for Hungarian, by Manetta (2006; 2011) for Hindi-Urdu, Todorović and Wurmbrand (2020) for Bosnian/Croatian/Serbian, Williams (2013) for English, by Angelopoulos (2019) for Greek, and by Kayne (2000) for a number of other languages. In essence, my treatment of *hogy* is somewhat similar to Wurmbrand’s (2001) treatment of German *zu*, which does not unambiguously indicate constituent-size, or Kim’s (2024) treatment of Korean complementizer *-ko*, which on its own does not add to the opacity of the embedded clause.

2.3 Cross-clausal movement across left-peripheries of various sizes

This subsection discusses which types of long movement can cross which embedded left-periphery positions. Particle movement is the most restricted as it cannot cross any of the overt left-periphery material introduced above. Focus movement is somewhat less restricted, and relative movement is the least restricted when it comes to long movement.

Particle extraction from the embedded clause to matrix Spec,TP is the most constrained cross-clausal operation. (That it can only proceed from subjunctive clauses is discussed in the next section.) Firstly, particle movement from relative clauses with a filled Spec,RelP is ungrammatical as in (31). Since the filled Spec,RelP implies the presence of a RelP layer in the embedded clause, (31) shows that particle movement to Spec,TP cannot cross RelPs. In general, throughout this section, crossing a *c-commanding* filled Spec,βP will be rephrased as crossing the *dominating* βP.

- (31) [TP × ... [VP ... [RelP relative pronoun [... XP ...]]]]
- ⟨_{PART} *Keresztül₁ lát-t-ad az ember-t, [[_{REL} aki] ⟨_{PART} keresztül₁ fut-ott-Ø
 across see-PST-2SG.DEF the man-ACC who across run-PST-3SG.INDF
 a város-on ____]
 the city-SUP
 Intended: ‘You saw a man who ran across the city.’

Secondly, if the Spec,ComP of the embedded clause is filled by adverbs like *következésképpen* ‘consequently’, no particle extraction can proceed to matrix Spec,TP in (32).⁸ That is, particle extraction to Spec,TP cannot proceed across ComPs.

- (32) [TP × ... [VP ... [ComP sentential adverbs [... XP ...]]]]
- ⟨_{PART} Keresztül₁ akar-t-am, [hogy (_{SADV} *következésképpen) fus-s-anak a város-on ____]
 across want-PST-1SG.DEF that consequently run-SUBJ-3PL.INDF the city-SUP
 ‘I wanted them to run across the city (*consequently).’

In a similar way, if the embedded clause has focused constituents, cross-clausal particle movement is blocked (Koopman and Szabolcsi 2000). Example (33) represents two versions for the embedded clause: In the a. version, there are two foci (both ‘with Mari’ and ‘the city’ are focused); in the b. version, there is a single focus (only ‘with Mari’ is focused). Particle movement to the matrix clause is ruled out in both versions. In other words, movement to matrix Spec,TP cannot cross embedded Foc1P and Foc2P.⁹

- (33) [TP × ... [VP ... ([Foc2P focus) [Foc1P focus [... XP ...]]]]]
- a. ⟨_{PART} *Keresztül₁ akar-t-am, [hogy [_{FOC} CSAK MARI-VAL] fus-s-anak ⟨_{PART} keresztül₁ [_{FOC}
 across want-PST-1SG.INDF that only Mari-INS run-SUBJ-3PL.INDF across
 CSAK A VÁROS-ON]₂ ____]
 only the city-SUP
 b. ⟨_{PART} *Keresztül₁ akar-t-am, [hogy [_{FOC} CSAK MARI-VAL] fus-s-anak ⟨_{PART} keresztül₁ a
 across want-PST-1SG.INDF that only Mari-INS run-SUBJ-3PL.INDF across the
 város-on ____]
 city-SUP
 Intended: ‘I wanted them to run across {the city/ ONLY THE CITY} ONLY WITH MARI.’

Finally, if the embedded clause contains another particle, no particle can move from within the embedded clause to the matrix clause i.e. movement to Spec,TP cannot proceed across TPs. In (34), the PP-internally generated *keresztül* ‘across’ cannot move to the matrix Spec,TP if the embedded clause has a particle like *el* ‘away’ base generated in Spec,TP. That is, particle movement to matrix Spec,TP is ungrammatical across TPs.

- (34) [TP × ... [VP ... [TP particle [... XP ...]]]]
- ⟨_{PART} Keresztül₁ akar-t-am, [hogy (_{PART} *el) fus-s-anak a város-on ____]
 across want-PST-1SG.INDF that away run-SUBJ-3PL.INDF the city-SUP
 ‘I wanted them to run across the city (*away).’

The next type of subextraction is focus/*wh*-movement to matrix Spec,FocP. Just like particle movement, focus/*wh*-movement in (35) cannot proceed from a clause with a filled embedded Spec,RelP, i.e. movement to matrix Spec,FocP is ruled out from RelPs.

⁸Subjunctive clauses are degraded with most sentential adverbs even without particle movement. The adverb *következésképpen* ‘consequently’ is more or less compatible with subjunctives because it is not an evaluative adverb.

⁹There are things to note for (33). Firstly, in the first version of (33), the phonologically heavier focus is scrambled after the particle (É. Kiss 2008). Secondly, if no long particle movement happens in the sentence, the matrix clause is more natural with the expletive *azt* in the particle position (see Section 3.2.2 for further discussion).

- (35) $\checkmark$ -----
 [FocP $\times$... [VP ... [RelP relative pronoun [... XP ...]]]]
 {_{Foc} CSAK A VÁROS-ON₂ lát-t-ad₁ _____ az ember-t, [[_{REL} aki] {_{Foc} CSAK A VÁROS-ON₂ }
 only the city-SUP see-PST-2SG.DEF the man-ACC who only the city-SUP
 fut-ott-Ø₃ [_{PART} keresztül]₄ _____
 run-PST-3SG.INDF across
 With low-clause foci: ‘You saw the man who had ran across ONLY THE CITY.’

Furthermore, (36) indicates that similarly to particle movement, focus movement cannot proceed if high sentential adverbs like *szerencsére* ‘fortunately’ or *sajnos* ‘unfortunately’ occur in the embedded clause’s Spec,Comp. That is, movement to matrix Spec,FocP is ruled out across CompPs.

- (36) $\checkmark$ -----
 [FocP $\times$... [VP ... [Comp sentential adverbs [... XP ...]]]]
 [_{Foc} (CSAK) A VÁROS-ON₂ hall-ott-am₁ _____, [hogy (_{SADV} ??szerencsére/ ??sajnos) [_{PART} keresztül]₃
 only the city-SUP hear-PST-1SG.DEF that fortunately unfortunately across
 fut-ott-ak _____
 run-PST-3PL.INDF
 ‘I heard that (??fortunately/??unfortunately) they ran across ONLY THE CITY.’

As discussed in Section 2.1, CompPs describe the content of speech events: Once we add *szerencsére* ‘fortunately’ to the complement clause of *hall*, it becomes the description of someone’s report heard rather than the event heard. Another way to elicit a Comp is to use a matrix verb whose complement clause is necessarily speech-reporting. For example, if we replace *hall* ‘hear’ by a manner-of-speech verbs like *morog* ‘growl’ and *hazudik* ‘lie’, we find a similar decrease in grammaticality for focus/*wh*-extraction in (37). In short, (37) is another example for the impossibility of focus/*wh*-movement from CompPs.¹⁰

- (37) $\checkmark$ -----
 [FocP $\times$... [VP manner-of-speech verb [Comp [... XP ...]]]]
 ??[_{Foc} Melyik város-on₂ {morog-t-ad/ hazud-t-ad}₁ _____, [hogy [_{PART} keresztül]₃ fut-nak
 which city-SUP growl-PST-2SG.INDF lie-PST-2SG.INDF that across run-PST-3PL.INDF
 _____₂ _____₃]?
 Intended: ‘Which city did you {growl/ lie} that they ran across?’

On the other hand, Horváth (1986) as well as É. Kiss (2002) mention that *wh*-phrases can move across other *wh*-phrases. In (38), even if the embedded has a focus, this does not block long focus movement to the matrix clause. Assuming the matrix landing site is Spec,Foc2P and the embedded focus is in Spec,Foc1P, focus movement to Spec,Foc2P can proceed from Foc1Ps.¹¹

- (38) $\checkmark$ -----
 [Foc2P $\checkmark$... [VP ... [Foc1P focus [... XP ...]]]]
 [_{Foc} CSAK A VÁROS-ON₁] akar-t-am₂ _____, [hogy [_{Foc} CSAK MARI-VAL] fus-s-anak₃
 only the city-SUP want-PST-1SG.DEF that only Mari-INS run-SUBJ-3PL.INDF
 [_{PART} keresztül]₄ _____₃ _____₁ _____₄]
 across
 ‘ONLY THE CITY I wanted them to run across ONLY WITH MARI.’

Finally, cross-clausal focus movement can proceed across particles in embedded Spec,TPs, i.e. across embedded TPs:

¹⁰Interestingly, *mond* ‘say’ does not behave like a manner of speech verb in this respect. One reason may be that *mond* can have both factive and non-factive complement clauses whereas other manner of speech verbs can only have non-factive ones (for a more sophisticated discussion, see de Cuba and Ürögdi 2009). In general, the central feature of Comp-sized clauses is that the person uttering the matrix clause distances themselves from the embedded proposition in some way, which is not necessary for *mond*, but necessary for manner of speech verbs. Particle extraction from the complement clauses of manner of speech verbs is not discussed separately as these clauses are just another kind of tensed clauses discussed in the next section.

¹¹Since moving a focus or *wh*-constituent across two other focused constituents raises the question whether there are more than two focus projections (e.g. Foc3P, Foc4P etc.), I am setting the problem aside.

- (39) $\downarrow$ [FocP ✓ ... [VP ... [TP particle [... XP ...]]]]
- [_{FOC} (CSAK) A VÁROS-ON]₂ hall-ott-am₁ __₁, [hogy [_{PART} keresztül]₃ fut-ott-ak __₂ __₃].
 only the city-SUP hear-PST-1SG.DEF that across run-3PL.INDF
 ‘I heard that they ran across ONLY THE CITY.’

The last type of long movement to be tested is the movement of relative pronouns to the matrix Spec,RelP. Similarly to the movement types above, relative movement cannot cross constituents in Spec,RelP i.e. movement to matrix Spec, RelP cannot proceed across RelPs:

- (40) $\downarrow$ [RelP × ... [VP ... [RelP relative pronoun [... XP ...]]]]
- *Ez az a város, [_{REL} amelyik-en]₁ lát-t-ad az ember-t, [hogy [_{PART} keresztül]₂ fut-ott-Ø __₁ __₂]
 this is the city which-SUP see-PST-2SG.DEF the man-ACC that across run-PST-3SG.INDF
 Intended: ‘This is the city which you saw the man who ran across.’

In contrast, cross-clausal relative pronoun movement is not blocked by either sentential adverbs occurring in the embedded Spec,ComP in (41), foci occurring in the embedded Spec,Foc1P or Foc2P in (42) (Horvath 1981), or particles occurring in the embedded Spec,TP in (42). Furthermore, (41) indicates that the ComP complements of manner of speech verbs are also transparent to relative extraction. In short, movement to Spec,RelP can cross ComPs, Foc1Ps, Foc2Ps, and TP_s.¹²

- (41) $\downarrow$ [RelP ✓ ... [VP ... [ComP sentential adverbs [... XP ...]]]]
- Ez az a város, [_{REL} amelyik-en]₁ az-t { hall-ott-am/ hazud-t-am/ morog-t-am}, [hogy
 this that the city which-SUP it-ACC hear-PST-1SG.DEF lie-PST-1SG.DEF growl-PST-1SG.DEF that
 (_{SADV} szerencsére) [_{PART} keresztül]₂ fut-ott-ak __₁ __₂]
 fortunately across run-PST-3PL.INDF
 ‘This is the city which I {heard/ lied/ growled} that they (fortunately) ran across.’

- (42) $\downarrow$ [RelP ✓ ... [VP ... [Foc2P focus [Foc1P focus [... [TP particle [... XP ...]]]]]]
- Ez az a város, [_{REL} amelyik-en]₁ az-t akar-t-am, [hogy [_{FOC} CSAK PETI] fus-s-on₃
 this that the city which-SUP it-ACC want-PST-1SG.INDF that only Peti run-SUBJ-3SG.INDF
 [_{PART} keresztül]₄ [_{FOC} CSAK MARI-VAL] __₃ __₁ __₄]
 across only Mari-INS
 ‘This is the city across which I wanted ONLY PETI to run across ONLY WITH MARI’

These observations are summarized in the table in (43). The general pattern is that lower matrix positions can be reached from fewer embedded clauses and fewer movement types can cross higher left-periphery positions. In particular, RelPs block any extraction types, ComPs block focus/*wh*-extraction and particle movement, and Foc1Ps and TP_s block particle movement. As discussed above, it is clear that Foc2Ps block particle movement but not relative movement, whereas focusing from Foc2Ps was set aside (because of the question on the number of FocP projections).

(43) **Opacity depending on the embedded left-periphery**

	Particle extraction	<i>wh</i> -/focus extraction	Relative extraction
Filled Spec,TP	×	✓	✓
Filled Spec,Foc1P	×	✓	✓
Filled Spec,Foc2P	×	?	✓
Filled Spec,ComP	×	×	✓
Filled Spec,RelP	×	×	×

¹²Although relative movement in (41) is degraded, most speakers find it better than focus movement in (36) on the basis of my informal survey mentioned in Footnote 1.

Even though the literature occasionally posits constraints on Hungarian movement types independently (e.g. for particle movement, see Farkas and Sadock 1989), I adopt the attitude of Starke (2001) and Abels (2012) by handling all movement types under a *single descriptive generalization*. This approach includes the idea that opacity and islandhood are not defined in absolute terms but are always relativized to individual movement types, e.g. relative clauses (RelPs) are not handled as absolute islands. Thus, on the basis of (43), I draw the generalization in (44) which will be defined as a version of the *Williams Cycle* (Williams 1974, 2003, 2013) in Section 2.5.¹³

- (44) **Size-dependent opacity:** Movement to Spec, β P cannot proceed across a dominating α P, where $\alpha \geq \beta$ in the functional sequence.

The generalization in (44) also applies to structures with multiple embedding. In (45), long movement of *keresztül* ‘across’ to matrix Spec,TP can proceed from a doubly embedded clause if neither of the embedded clauses have any overt left-periphery material, that is, if neither of the embedded clauses are TP or larger. (In the next section, the size of clauses smaller than TP will be defined as sModP.) However, if Spec,FocP (either Foc1P or Foc2P) is filled in either the deepest or the intermediate clause, the structure is ungrammatical.

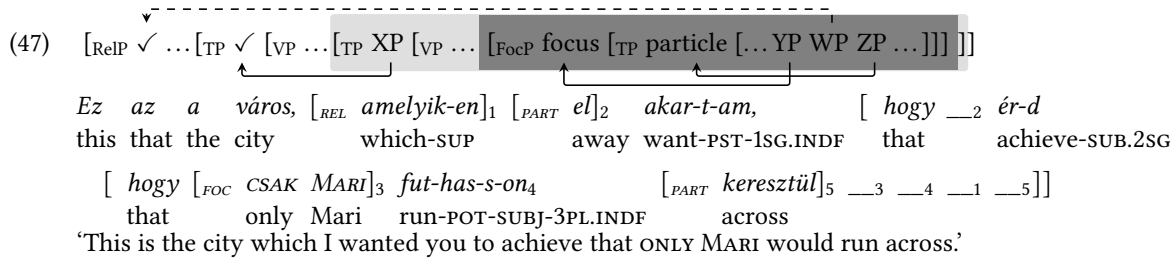
- (45) $[TP \times \dots [VP \dots [FocP \langle focus \rangle [\dots [VP \dots [FocP \langle focus \rangle [\dots XP \dots]]]]]]]]$
- $[_{PART} \text{ Keresztül}]_1 \text{ kell-ett-}\emptyset, \quad [\text{ hogy } \langle_{Foc} *CSAK \text{ MARI-VAL} \rangle \text{ akar-j-am,}$
 across be.needed-PST-3SG that only Mari-INS want-SUBJ-1SG.INDF
 $[\text{ hogy } \langle_{Foc} *CSAK \text{ MARI-VAL} \rangle \text{ fut-s-anak} \quad a \quad város-on \text{ } _1]]$
 that only Mari-INS run-SUBJ-3PL.INDF the city-SUP
 ‘I needed to want them to run across the city (*ONLY WITH MARI).’

In a similar way, even if focus movement (to either Spec,Foc1P or Spec,Foc2P) can proceed across multiple clauses, it gets blocked by intervening high sentential adverbs occurring in either of the clauses:

- (46) $[FocP \times \dots [VP \dots [ComP \text{ sentential adverbs } \dots [VP \dots [ComP \text{ sentential adverbs } \dots [XP]]]]]]$
- $[_{Foc} (CSAK) A \quad VÁROS-ON]_2 \text{ gondol-t-am}_1 \quad _1, [\text{ hogy } \langle_{SADV} ??szerencsére \rangle \text{ hall-ott-ad}$
 only the city-SUP think-PST-1SG.INDF that fortunately hear-PST-1SG.DEF
 $[\text{ hogy } \langle_{SADV} ??szerencsére \rangle [_{PART} \text{ keresztül}]_3 \text{ fut-ott-ak} \quad _2 \quad _3]$.
 that fortunately across run-PST-3PL.INDF
 ‘I thought that you (??fortunately) heard that (??fortunately) they ran across ONLY THE CITY.’

On the other hand, a single sentence may exhibit multiple movement types crossing each other if they all respect (44). For example, consider the double embedding structure in (47). Within the deepest embedded clause, there are two instances of movement: focus movement indicated by Index 3, and particle movement indicated by Index 5. There is additional particle movement (Index 2) from the intermediate embedded clause to the matrix clause. This particle movement (Index 2) is not interfered with by the previous two movement operations (Index 3 and 5) as those happen within the deepest embedded clause, that is, the particle movement (Index 2) does not cross any particles, foci, adverbs, or relative pronouns. In other words, (47) demonstrates that only crossed constituents lead to intervention effects. Finally, the relative movement of *amelyiken* ‘which’ (Index 1) originates in the deepest clause and lands in the matrix clause. This movement crosses *keresztül* ‘across’ in the Spec,TP and *csak Mari* ‘only Mari’ in Spec,FocP of the deepest clause, and *el* ‘away’ in the Spec,TP of the highest clause. These crossings are not problematic because both Foc and T are lower in *fseq* than Rel. (In addition, they are subjunctive clauses as discussed in Section 2.4.)

¹³In the spirit of Abels (2012), (44) could be re-phrased in terms of linear precedence or asymmetric c-command: Movement to the X matrix position across a Y embedded position is possible iff X c-commands Y within clauses. For instance, cross-clausal particle movement is blocked by any left-periphery operator as particles do not asymmetrically c-command any other operator. Foci c-command particles but are c-commanded by relative pronouns within clauses, so cross-clausal focus movement is blocked by relative pronouns but not by particles.

- (47) 

 Ez az a város, [REL amelyik-en]₁ [PART el]₂ akar-t-am, [hogy ₂ ér-d

 this that the city which-SUP away want-PST-1SG.INDF that achieve-SUB.2SG

 [hogy [FOC CSAK MARI]₃ fut-has-s-on₄ [PART keresztül]₅ ₃ ₄ ₁ ₅]

 that only Mari run-POT-SUBJ-3PL.INDF across

 ‘This is the city which I wanted you to achieve that ONLY MARI would run across.’

Successive cyclicity and intermediate stops are ignored for this example. In my proposal in Section 5, I give a principled explanation for the absence of successive cyclic stops via Hungarian clause edges. Therefore, even though the particle movement indicated by Index 2 as well as the relative movement indicated by Index 1 cross the boundary of the intermediate clause, the number of edge positions will not be a problem in (47).

2.4 The opacity of clause types

The extent to which an embedded clause is opaque is also dependent on its *type*. I will compare two Hungarian clause-types: subjunctive vs. tensed. I will trace back this difference to a difference in structural size: minimally sModP vs minimally TP, respectively. In this way, the opacity differences between clause types will be comparable to the opacity differences shown in the previous section.

Matrix verbs select for various types of embedded clauses, where the type of a clause is defined by the inflection of its verb. If the matrix verb *akar* ‘want’ occurs with a *that*-clause complement, this clause must be headed by subjunctive verbs rather than tensed ones (here past tense) as in (48) (Farkas and Sadock 1989).¹⁴ Importantly, Hungarian subjunctive inflection does not exhibit tense-contrasts, but only Agree with the subject and the object. Additionally, there is an edge marker *hogy* ‘that’ even in subjunctive clauses.

- (48) *Akar-t-am,* [Subjunctive *hogy Mari* {*vág-j-a/* **vág-t-a*} *fel az almá-t*]

 want-PST-1SG.DEF that Mari cut-SUBJ-3SG.DEF cut-PST-3SG.DEF up the apple-ACC

 ‘I wanted Mari to cut the apple.’

Other verbs that require subjunctive complement clauses are *kell* ‘it is needed that...’, *szabad* ‘it is allowed that ...’, *hagy* ‘let do that...’, *szeretne* ‘would like’.

In contrast, verbs like *hall* ‘hear’ select for embedded clauses which are headed by tensed verbs:

- (49) *Hall-ott-am,* [Tensed *hogy Mari* {*vág-t-a/* **vág-j-a*} *fel az almá-t*]

 hear-PST-1SG.DEF that Mari cut-PST-3SG.DEF cut-SUBJ-3SG.DEF up the apple-ACC

 ‘I heard that Mari was cutting the apple.’

Further verbs selecting for tensed clauses are *lát* ‘hear’, *hisz* ‘believe’, and *gondol* ‘think’. Manner of speech verbs like *hazudik* ‘lie’, *morog* ‘growl’, and *rikolt* ‘shout’ also select for tensed clauses, but these are at least ComPs.

Here I follow the intuition of Wurmbrand (2001; 2015), Ramchand and Svenonius (2014), and Wurmbrand and Lohninger (2020) in that I derive clause-type differences from the syntactic hierarchy. Hungarian subjunctive inflection does not show any contrasts for tense i.e. the morphosyntax does not have the past subjunctive vs. present subjunctive distinction. Furthermore, according to Koopman and Szabolcsi (2000), subjunctive complements pattern with infinitival complements in a number of respects even if subjunctive clauses are different from infinitival clauses in that they exhibit Agree with both the object and the subject (48). The subjunctive clauses discussed here are complements of modal raising and control verbs like *akar* ‘want’ and *kell* ‘is needed’ and alternate with infinitivals. Based on these, I assume that Hungarian subjunctive clauses can be categorized as Balkan-style reduced subjunctives (É. Kiss 2022), which are structurally smaller compared to indicatives in that they lack the syntactic encoding of tense (for cross-linguistic arguments on the reduced size of Balkan-style subjunctives, see Noonan 2007; Lohninger and Wurmbrand 2020; Wurmbrand and Lohninger 2023).

¹⁴Example (48) as it stands has an emphasis ‘I did want...’. A more neutral structure can be reached the insertion of the clausal expletive ‘azt’, which is discussed in Section 3.2.2.

More specifically, since the complementizer *hogy* does not contribute to clause size, bare subjunctive clauses lack the TP layer and are sModPs (for subjunctive modality) as in (26), whereas bare tensed are at least TPs as in (25). In *fseq*-terms, the T head is higher than the sMod head.¹⁵

(50) [VP *akar* [sModP *hogy* [sMod [VP ...]]]]

(51) [VP *hall* [TP *hogy* [T [sModP ...]]]]

Long particle extraction is possible from (certain) subjunctive clauses (É. Kiss 1994; Koopman and Szabolcsi 2000; Egressy to appear), specifically, from the complement clauses of modal and aspectual light verbs as in (52), but it is impossible from any tensed indicative clauses as in (53). If tense inflection is encoded in TP, which subjunctives but not indicatives lack, particle movement to Spec,TP cannot proceed across TPs.¹⁶

(52) [TP ✓ ... [VP *akar* [... V_{Subjunctive} [... XP ...]]]]
 [PART *Keresztül*]₁ *akar-t-am*, [Subjunctive *hogy fus-s-anak a város-on* ____]
 across want-PST-1SG.DEF that run-SUBJ-3PL.INDF the city-SUP
 ‘I wanted them to run across the city.’

(53) [TP × ... [VP *hall* [... V_{Tensed} [... XP ...]]]]
 ??[PART *Keresztül*]₁ *hall-ott-am*, [Tensed *hogy* [PART *keresztül*]₁ *fut-ott-ak a város-on* ____]
 across hear-PST-1SG.DEF that across run-PST-3PL.INDF the city-SUP
 Intended: ‘I heard that they ran across the city.’

The opacity-difference between clause-types does not occur for focus and relative movement. Firstly, cross-clausal movement landing in matrix Spec,FocP is grammatical from subjunctive as well as tensed clauses. The examples below demonstrate this with long-*wh*-movement.

(54) [FocP ✓ ... [VP *akar* [... V_{Subjunctive} [... XP ...]]]]
 [FOC *Melyik város-on*]₁ *akar-t-ad* ____₂, [Subjunctive *hogy* [PART *keresztül*]₃ *fus-s-anak* ____₁ ____₃]?
 which city-SUP want-PST-2SG.INDF that across run-SUBJ-3PL.INDF
 ‘Which city did you want them to run across?’

(55) [FocP ✓ ... [VP *hall* [... V_{Tensed} [... XP ...]]]]
 [FOC *Melyik város-on*]₁ *hall-ott-ad* ____₂, [Tensed *hogy* [PART *keresztül*]₃ *fut-ott-ak* ____₁ ____₃]?
 which city-SUP hear-PST-2SG.DEF that across run-PST-3PL.INDF
 ‘Which city did you hear that they ran across?’

Secondly, relative extraction is also possible from both clause types as in (56) and (57).

(56) [RelP ✓ ... [VP *akar* [... V_{Subjunctive} [... XP ...]]]]
Ez az a város, [REL *amelyik-en*]₁ *az-t akar-t-ad*,
 this that the city which-SUP it-ACC want-PST-2SG.INDF
 [Subjunctive *hogy* [PART *keresztül*]₂ *fus-s-anak* ____₁ ____₂]
 that across run-SUBJ-3PL.INDF
 ‘This is the city which you wanted them to run across.’

¹⁵An anonymous reviewer points out that heads responsible for mood and modality are high Cinque’s hierarchy (Cinque 1999). However, heads for necessity are lower than tense heads, and the Hungarian subjunctive has a necessity meaning in isolation.

¹⁶An anonymous reviewer mentions that not all Hungarian subjunctive clauses are transparent to long-particle movement. I need to emphasize that all I claim is that particle extraction is *never* possible from indicative TPs, whereas it *may* be possible from subjunctive clauses. Whether different matrix verbs can select for subjunctive clauses of different sizes is a problem I am setting aside.

- (57) $\checkmark$ -----
 [RelP $\checkmark$... [VP *hall* [... V_{Tensed} [... XP ...]]]]
Ez az a város, [REL amelyik-en]₁ hall-ott-ad, [Tensed hogy [PART keresztül]₂ fut-ott-ak ____₁ ____₂]
 this is the city which-SUP hear-PST-1PL.INDF that across run-PST-1PL.INDF
 ‘This is the city which you heard that they ran across.’

In sum, setting aside overt left-periphery in the embedded clause, the only movement type that the presence of the TP layer of the tensed indicative clauses blocks is particle movement to Spec,TP. That is, whether the presence of the TP layer is indicated by the embedded inflection as in (53) or the presence of a particle in the embedded Spec,TP as in (34), we are dealing with the same instance of size-dependent opacity: movement to Spec,TP cannot proceed across TPs. In short, the opacity-difference between clause-types falls under the generalization repeated in (58).

- (58) **Size-dependent opacity:** Movement to Spec, β P cannot proceed across a dominating α P, where $\alpha \geq \beta$ in the functional sequence.

Based on the previous section, (58) might have been reduced to a generalization on blocking effects between c-commanding operators, e.g. foci block cross-clausal particle movement because particles are c-commanded by foci within clauses.¹⁷ However, the opacity of tensed TP clauses to particle movement does not have a straightforward c-command-based translation. In short, the clause-type data supports the domination-based phrasing of (58).

The discussion so far has been only concerned with the effect of clause-types in isolation i.e. how bare indicative TPs are more opaque than bare subjunctive sModPs. Above the sModP and TP projections encoding clause types, embedded clauses can have more or less additional left-periphery projections. For instance, *akar* ‘want’ has been shown to select for a subjunctive clauses, but subjunctive sModPs turn into FocPs with the addition of foci in (59). The addition of left-periphery material has the expected effect on the embedded clause’s opacity: While sModP subjunctive clauses are transparent to particle extraction in (52) because T>sMod in *fseq*, the addition of foci makes long particle movement impossible in (59) because Foc>T.

- (59) $\checkmark$ -----
 [TP $\times$... [VP ... [FocP focus [... [sModP V_{Subjunctive} ... XP ...]]]]]]
 (_{PART} *Keresztül)₁ akar-t-am, [hogy [_{FOC} CSAK MARI-VAL] fus-s-anak₂ (_{PART} keresztül)₁ ____₂
 across want-PST-1SG.INDF that only Mari-INS run-SUBJ-3PL.INDF across
 a város-on ____]
 the city-SUP
 ‘I wanted them to run across the city ONLY WITH MARI.’

Another example of this interaction is (60). The addition of a focus to the embedded tensed TP yields a FocP. However, this does not affect the clause’s transparency to relative extraction because Rel>Foc>T in *fseq*.

- (60) $\checkmark$ -----
 [RelP $\checkmark$... [VP ... [FocP focus [... [TP V_{Tensed} ... XP ...]]]]]]
Ez az a város, [REL amelyik-en]₁ hall-ott-ad, [hogy [_{FOC} CSAK MARI-VAL] fut-ott-ak
 this that the city which-SUP hear-PST-1SG.DEF that only Mari-INS run-PST-1PL.INDF
 [_{PART} keresztül]₂ ____₁ ____₂]
 across
 ‘This is the city which you heard that they ran across ONLY WITH MARI.’

Hence, what clause-type really defines is the *minimal* size of the clause: In order to be [+tense], a clause needs to contain a T head, i.e. to be at least TP. Put differently, the selectional specification of the matrix verbs is the following: While verbs like *akar* ‘want’ select for [-tense] clauses¹⁸(at least sModP), verbs like *hall* ‘hear’ select for [+tense] clauses (at least TP). For non-minimal complements (e.g. FocPs and ComPs), the tense feature percolates up to the maximal projection of the embedded clause in the spirit of Grimshaw (2000).

¹⁷For accounts using *Relativized Minimality* (Rizzi 1990) to account for *c-command-based intervention*, see (Starke 2001; Abels 2012).

¹⁸In this analysis, [-tense] clauses emerge in two ways: 1) with a [-tense] T head; 2) in the absence of T head. This can be encoded by stating that subjunctive-selecting verbs are incompatible with [+tense] clauses.

Finally, the verb's position and the morpheme order within verbs has not been addressed. The verb is assumed to move to sMod in sModP clauses, to T in TP clauses, and to Foc clauses with a focus. Within verbs, the subjunctive and tense morphemes both occur between the verb root and the agreement suffix. Why the agreement suffix gets realized behind every other morpheme in the verb (i.e. high in the structure) is not addressed in this paper.

2.5 Interim summary: Williams Cycle in Hungarian

While the section has been restricted to size-dependent opacity in Hungarian, this subsection demonstrates that the generalization drawn corresponds to a version of the *Williams Cycle* (Williams 1974, 2003, 2013) in (61). For now, the Williams Cycle is a descriptive generalization; the theories deriving the Williams Cycle are discussed in Sections 4 and 5.

- (61) **Williams Cycle:** Movement to Spec, β P cannot proceed from Spec, α P or across α P, where $\alpha > \beta$ in the functional sequence. (based on Poole 2023; Williams 2003)

The Williams Cycle has been shown to hold in some form in a number of languages including German (Müller 2014b,a), Swahili (Meadows 2023), Mandarin Chinese (Meadows and Yan 2024), and Georgian (Bondarenko 2024). The best-known illustration of the Williams Cycle is the ban on *hyper-raising* in English (originally observed by Chomsky 1973). For instance, consider (62) in which *John* cannot raise from the embedded CP to the matrix Spec,TP. Under (61), the ungrammaticality of (62) is expected as C>T in *fseq*.

- (62) *John₁ seems [CP__₁ has eaten an apple]

In contrast, subject raising can proceed from embedded TPs:

- (63) John₁ seems [TP__₁ to have eaten an apple]

The impossibility of raising to matrix Spec,TP from embedded CPs falls under the Williams Cycle in (61) if C>T in *fseq*.

There is only one difference between the Williams Cycle in (61) and the generalization on Hungarian size-dependent opacity repeated in (64): While (61) does not rule out movement to Spec, α P across α P, (64) does. Therefore, (64) can be referred to as the *Strict Williams Cycle*. For Hungarian, this restriction is necessary because relative movement to Spec,RelP cannot escape RelPs and particle movement to Spec,TP cannot cross TPs.

- (64) **Size-dependent opacity/Strict Williams Cycle:** Movement to Spec, β P cannot proceed across a dominating α P, where $\alpha \geq \beta$ in the functional sequence.

Patterns describable with the Strict Williams Cycle in (64) have been attested in other languages, too. For example, as Abels (2012) observes, the movement of Italian relative pronouns is possible from clauses with a focus or *wh*-phrase as in (65) (Rizzi 1980:p 50) but focusing or *wh*-movement is not possible from clauses with a relative pronoun as in (66) (Cinque 1991:p 28). If we assume that focusing and *wh*-movement both land in Spec,FocP, relative movement lands in Spec,RelP, and Rel>Foc in the Italian *fseq* (Rizzi (1997); Rizzi uses Force instead of Rel), the grammaticality of (65) and the ungrammaticality of (66) is expected even under the less restrictive Williams Cycle version in (61).

- (65) [RelP ✓ ... [VP ... [FocP focus [... XP ...]]]]

Tou fratello, [REL a cui]₁ mi domando, [[FOC che storie] abbiamo raccontato __₁], era molto
 your brother to whom I wonder which stories have told was very
preoccupato
 troubled
 'Your brother, to whom I wonder which stories they told was very troubled.'

- (66) $[\text{FocP} \times \dots [\text{VP} \dots [\text{RelP} \text{ relative pronoun } [\dots \text{XP} \dots]]]]$
- * $[\text{Foc} \text{ A } \text{chi}]_1 \text{ hai conosciuto qualcuno, } [[\text{REL} \text{ che}] \text{ follesse parlare } __1]$
 to whom have met someone who wanted to.speak
 Intended: ‘Who have you met someone who wanted to talk to?’

However, Italian *wh*-movement cannot cross foci as in (67) (Abels 2012:p. 241). That is, if we assume that both focusing and *wh*-movement land in Spec,FocP, the ungrammaticality of (67) is unexpected under (61). Instead, similar to Hungarian, Italian seems to exhibit the Strict Williams Cycle in (64): Movement to Spec,FocP cannot proceed across FocPs. (Notice that while the multiple FocPs in Hungarian allowed foci escaping FocIP, Italian does not tolerate multiple foci/*wh*-words within clauses (Rizzi 1980).)

- (67) $[\text{FocP} \times \dots [\text{VP} \dots [\text{Foc} \text{ wh-constituent } [\dots \text{XP} \dots]]]]$
- * $[\text{Foc} \text{ A } \text{chi}]_1 \text{ pensi } [\text{che } [\text{Foc} \text{ QUESTO}] \text{ abbiamo detto } __1]$
 to whom think that this have said
 Intended: ‘Who do you think they said THIS?’

The exact distribution of the more and less restrictive Williams Cycle versions in (61) and (64) is a question for further research. As of now, it seems that (61) and (64) are both attested in human language: While languages like English exhibiting (61) allow unbounded *successive cyclic* movement from CP to Spec,CP or TP to Spec,TP (e.g. Chomsky 1973, 2001), the data from Hungarian and Italian fall under this stricter version in (64). My proposal in Section 5 will be concerned with the derivation of the strict Williams Cycle in (64) and will argue that (64) arises in Hungarian because of the absence of successive cyclic movement in the clausal domain.

3 Position-dependent opacity in Hungarian

The opacity of embedded clauses does not only depend on their size but also their position in the matrix clause. Section 3.1 introduces the concept of *clausal expletives* and the idea that they are base-generated in a single constituent with the embedded clause. Section 3.2 focuses on the positions in which clausal expletives can show up but the clauses associated with them cannot. Section 3.3 introduces the first part of position dependent data; the possibility of the various subextraction types from embedded clauses moved to the left-periphery. Since the clauses themselves are banned from most left-periphery positions, Section 3.4 supplements the position-dependent discussion with data involving subextraction possibilities constrained by various expletive-positions.

3.1 Embedding with expletives

Similarly to a number of languages, *that*-clauses in Hungarian co-occur with overt or covert pronominals (e.g. É. Kiss 1987; Kenesei 1994; Horvath 1997). English occasionally exhibits this behaviour as well e.g. *I doubt it that you come* (Kiparsky and Kiparsky 1971). The Hungarian pronominal *az* ‘it’ associated with clauses bears the case suffix assigned by the matrix verb. I will refer to *az* ‘it’ with the term (*clausal*) *expletive* following Kenesei (1994), Horváth (1997), É. Kiss (2002), and Den Dikken (2018).

For example, *hall* ‘hear’ assigns accusative case to the expletive, turning it into *az-t* ‘it-ACC’ in (68). The expletive can stay VP-internal or occur in various left-periphery positions. VP-internally, expletives must undergo *pro*-drop (e.g. the post-verbal expletive in (68)), whereas expletives remain overt if they occur in the left-periphery (e.g. the pre-verbal expletive in (68); see below detailed discussion on the effects of overt expletives). Essentially, Hungarian complement *that*-clauses are always embedded with an expletive (Kenesei 1994). One piece of evidence for the general presence of expletives is that the *pro*-dropped accusative expletive still triggers definite agreement on the verb in (68) (É. Kiss 2002).

- (68) $[\langle \text{expletive}_{\text{IACC}} \rangle \dots [\text{VP } \text{hall}_{\text{DEF}} [\langle \text{expletive}_{\text{IACC}} \rangle [\text{hogy} \dots]_i]]$
- $\langle \text{Az-t} \rangle \text{ hall-ott} \{ \text{-ad} / \text{-}^* \text{-} \text{ál} \} \quad \langle \text{az-ACC} \rangle, [\text{hogy } [\text{PART } \text{keresztül}]_1 \text{ fut-ott-am} \quad \text{a város-on } __1]$
 it-ACC hear-PST-2SG.DEF/-2SG.INDF pro.ACC that across run-PST-1SG.INDF the city-SUP
 ‘You heard that I ran across the city.’

Finally, even if the small clause remains VP-internal and the clausal expletive undergoes *pro*-drop, the embedded clause may undergo movement to a FocP- or ComP-sister position:

- (73) [✓ [ComP sentential adverbs[✓ [FocP ... [VP V [SC az [hogy ...]]]]]]]]]
 ↳
 <_{TOP} hogy Mari_j ismer-t-e Péter-t₁ [_{SADV} szerencsére] <_{TOP} hogy Mari_j ismer-t-e
 that Mari know-PST-3SG.DEF Péter-ACC fortunately that Mari know-PST-1SG.DEF
 Péter-t₁ [_{FOC} CSAK ő_{i/??j}] mond-t-a [_{SC} pro —₁]
 Péter-ACC only she say-PST-3SG.DEF pro.ACC
 ‘As for that Mari_j had known Péter, fortunately, she_{i/??j} did not tell it.’

On the basis (71), (72), and (73), I draw the generalization in (74): After being base generated in the small clause, the embedded clause may be moved out but at every derivational step, it must c-command its expletive. The reason for this may be that the expletive is pronominal in nature.

- (74) A clausal expletive must be bound by a c-commanding clause associated with it at each derivational step.

3.2 Where no full clauses are tolerated

The previous subsection established that every embedded clause is base generated in a VP-internal small clause with an expletive. After this, the embedded clause *can* undergo movement with or without the rest of the small clause to certain positions. This section discusses the positions in which the embedded clause *must not* stay. Since clauses and expletives are generated as one constituent and the clause needs to c-command the expletive at every step, the positions where the expletive but not the clause can occur will be assumed to be the positions from which the embedded clause has moved out.

3.2.1 No full clauses can remain VP-internal

As proposed in Section 3.1, only VP-internal expletives can be *pro*-dropped. For example, in (75), the sublativ case marker left behind after the *pro*-drop of the expletive indicates that the expletive is VP-internal. The adjuncts *tegnap* and *egy buliban* obligatory occurring between the expletive verb and post-verbal embedded clause suggest that the embedded clause must undergo obligatory extraposition to the right (É. Kiss 2002).

- (75) [[[VP V [hogy ...]] adjuncts] ✓]
 ↳
 [_{FOC} CSAK ő_{i/*j}] hivatkoz-ott-Ø az-rá —₁ < tegnap> < egy buli-ban>, [hogy Mari_j alsz-ik]₁
 only she refer-PST-3SG.INDF pro-SUB yesterday a party-IN that Mari sleep-3SG.INDF
 (* tegnap) (* egy buliban).
 yesterday a party-IN
 ‘Fortunately, only she_{i/*j} referred (to the fact) yesterday in a party that Mari_j was sleeping’

Based on examples like (75) and further examples involving noun-complement clauses, É. Kiss’s (2002) draws the generalization in (76), which is interpreted here as a syntactic constraint. For this paper, the relevant consequence of (76) is that tensed and subjunctive clauses introduced by *hogy* ‘that’ cannot remain VP-internal. The syntactic and semantic reasons behind clausal extraposition have been discussed extensively in the cross-linguistic literature (e.g. Moulton 2015; Bruening 2018). (Where clauses land after rightward movement is discussed below and in Section 5.)

- (76) **That-clause Constraint:** A *that*-clause cannot be internal to a lexical projection. (É. Kiss 2002:p. 237)

In the grammatical examples (71), (72), and (73), the clause or the small clause containing the clause landed in a position left-attached to the FocP or ComP projections. These examples indicate that (76) can be also satisfied by the leftward movement of the clause. Additionally, the rightward movement of the embedded clause to a post-verbal position can take place from the FocP- or ComP-sister small clause as in (77). The adjuncts showing up before the embedded clause indicate that (71) does not violate (76) by a VP-internal clause.

- (77) [[SC *az [hogy ...]*] [ComP sentential adverbs [[SC *az [hogy ...]*] [FocP ... [VP V SC]] ...] ✓
- ⟨_{TOP,SC} *Az-t* ____₂ ⟩₁ [_{SADV} *szerencsére*] ⟨_{TOP,SC} *az-t* ____₂ ⟩₁ [_{FOC} *CSAK Ő_{i/??j}*] *mond-t-a* ____₁ *tegnap*
 it-ACC fortunately it-ACC only she say-PST-3SG.DEF yesterday
egy buli-ban [hogy Mari_j ismer-t-e Péter-t]₂
 a party-IN that Mari know-PST-3SG.DEF Péter-ACC
 ‘As for that Mari_j had known Péter, fortunately, ONLY SHE_{i/??j} did not tell it.’

3.2.2 No clauses can be contained in a specifier dominated by some FocP

Since, full finite clauses are banned from left-periphery specifiers dominated by FocPs, if the small clause moves to such specifier positions, the embedded clause must be immediately moved out of it.

Firstly, É. Kiss (1981) observed that entire clauses cannot be foci, the movement of clauses cannot land in Spec,FocP. Thus, in (78), neither the embedded clause nor the small clause containing the clause and the expletive can occur in Spec,FocP. Furthermore, the fact that the presence of another focus does not make the sentence better shows that embedded clauses cannot occur either in Spec,Foc1P or Spec,Foc2P.

- (78) [_{Foc2P/Foc1P} × [... [VP V [*hogy ...*]]]]]
- * (_{FOC} *CSAK EMMÁ-NAK*) [_{FOC} *CSAK (AZ-T) HOGY ISMER-T-EM PÉTER-T*]₁ *mond-t-am* (_{FOC} *CSAK*
 only Emma-DAT only it-ACC that know-PST-1SG.DEF Péter-ACC say-PST-1SG.DEF only
EMMÁ-NAK) ____₁.
 Emma-DAT
 Intended: ‘I only told (TO EMMA) THAT I HAD KNOWN PÉTER.’

Still, the embedded clause can get a focus-interpretation if the clause leaves the small clause Spec,FocP in a way that only the overt expletive is left behind. In the a. version of (79), the small clause occurs in Spec,Foc2P; in the b. version, the small clause occurs in Spec,Foc1P. Either of these are grammatical if the embedded clause re-merges to the left of FocP or ComP or in the right-periphery. (Notice that the embedded clause takes different scopes in the two different positions.)

- (79) [_{Foc1P/Foc2P/ComP} ✓ [_{Foc1P/Foc2P} [SC *az [hogy ...]*] [... [VP V SC]] ...] ✓
- a. ⟨ *Hogy ismer-t-em Péter-t* ⟩₂ [_{FOC} *CSAK AZ-T* ____₂]₁ *mond-t-am* [_{FOC} *CSAK EMMÁ-NAK*]
 that know-PST-1SG.DEF Péter-ACC only it-ACC say-PST-1SG.DEF only Emma-DAT
 ____₃ ____₁ ⟨ *hogy ismer-t-em Péter-t* ⟩₂.
 that know-PST-1SG.DEF Péter-ACC
 ‘The only thing such that I told it only to Emma was that I know Peter.’
- b. ⟨ *Hogy ismer-t-em Péter-t* ⟩₂ [_{FOC} *CSAK EMMÁ-NAK*] *mond-t-am*₃ [_{FOC} *CSAK AZ-T* ____₂]₁
 that know-PST-1SG.DEF Péter-ACC only Emma-DAT say-PST-1SG.DEF only it-ACC
 ____₃ ____₁ ⟨ *hogy ismer-t-em Péter-t* ⟩₂.
 that know-PST-1SG.DEF Péter-ACC
 ‘Emma was the only person such that the only thing I told them was that I know Peter.’

Embedded clauses cannot occur in Spec,TP, either. A subset of Hungarian verbs including *akar* ‘want’ counts as *stress-avoiding*: These cannot bear the phonetically neutral sentential stress, that is, they cannot be sentence-initial unless they are contrastively or emphatically stressed as in (48) (Komlósy 1992; Szendrői 2003). Particles can move to Spec,TP to bear neutral sentential stress as in (13), however, entire clauses or small clauses containing the clause cannot behave like particles as in (80).

- (80) $[TP \times [\dots [VP \ V \ [hogy \dots]]]]]]$
 $\uparrow$
 $*[PART \ (Az-t,) \ hogy \ [PART \ keresztül]_2 \ fu-s \quad \quad \quad a \ város-on \ _2]_1 \ akar-t-am \ _1$
 it-ACC that across run-SUBJ.2SG.INDF the city-SUP want-PST-1SG.INDF
 Intended: 'I wanted you to run across the city.'

Nevertheless, when the embedded clause leaves the small clause in Spec,TP (in a way that the expletive is left behind) and re-merges in the right-periphery as in (81), the sentence is grammatical. The remnant of the small clause in Spec,TP works like any other particle, e.g. it becomes post-verbal if the verb undergoes movement to the Foc head in (82). Since VP-internal expletives must undergo *pro*-drop, any post-verbal expletive that is overt but non-focused is assumed to be contained in Spec,TP. Notice that the post-verbal clause in (82) is still moved to a position above adjuncts in the right periphery.

- (81) $[FocP/Comp \ \checkmark \ [TP \ [SC \ az \ [hogy \dots]] \ [\dots [VP \ V \ SC] \dots \checkmark]$
 $\uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow$
Peti $[PART \ az-t \ _3]_1 \ akar-t-a \ _1,$ $[\ hogy \ [PART \ keresztül]_2 \ fu-s \quad \quad \quad a \ város-on \ _2]_3$
 Peti it-ACC want-PST-3SG.INDF that across run-SUBJ.2SG.INDF the city-SUP
 'Peti wanted you to run across the city.'

- (82) $[FocP \ focus \ Foc+\checkmark \ [TP \ [SC \ az \ [hogy \dots]] \ T+V \ [\dots [VP \ V \ SC] \dots \checkmark]$
 $\uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow$
 $[Foc \ CSAK \ PETI] \ akar-t-a_4 \quad \quad \quad [PART \ az-t \ _3]_1 \ _4 \ _1 \ tegnap \ egy \ buli-ban, \ [\ hogy \ [PART$
 only Peti want-PST-3SG.INDF it-ACC yesterday a party-IN that
 $keresztül]_2 \ fu-s \quad \quad \quad a \ város-on \ _2]_3$
 across run-SUBJ.2SG.INDF the city-SUP
 'Yesterday at the party, ONLY PETI wanted you to run across the city.'

The constraint in (83) summarizes these facts by claiming that while embedded clauses are tolerated as sisters to maximal FocP Comp projections, embedded clauses must be moved out from Spec,TP, Spec,Foc1P, and Spec,Foc2P. In addition, É.Kiss 2002:p.88-89 mentions (83) also holds for DP-internal relative clauses, e.g. relative clauses move out of focused DPs. Kenesei (1994) proposes that the ban on left-periphery clauses within FocP has a phonological explanation, which is compatible with the assumption that the trace of embedded clauses can still in these positions. Here I will remain agnostic about the exact motivation of (83).

- (83) No specifier dominated by FocP (either Foc1P or Foc2P) can contain finite (subjunctive or tensed) embedded clauses.

Both the ban on VP-internal clauses in (76) and the ban on clauses in FocP-dominated specifiers in (83) are satisfied by the movement of clauses to the right-periphery or to the left-periphery. While the exact landing sites are discussed as a part of the proposal in Section 5, examples like (84) and (85) show that the timing of leftward movement is determined by syntactic factors. The matrix clauses in these examples contain two small clauses: one in Spec,TP and another in Spec,FocP. While the two small clauses are the relative and subrelative arguments of *következtet* 'infer' in both examples, the main clause in (84) has the relative small clause in Spec,FocP and the subrelative small clause in Spec,TP; in (85), the roles are flipped: the subrelative small clause is in Spec,FocP, and the relative small clause is in Spec,TP. Similar examples are provided by É. Kiss (2002:p.238).

- (84) $[Foc \ CSAK \ AB-BÓL \ _2] \ következtet-t-em_1 \ [PART \ ar-ra \ _3] \ _1 \ tegnap \ [\ hogy \ Mari \ haza \ men-t-\emptyset]_3$
 only it-EL infer-PST-1SG.DEF it-SUB yesterday that Mari home go-PST-3SG.INDF
 $[\ hogy \ nem \ vol-t-\emptyset \quad \quad \quad elég \ lány \ a \ buli-ban]_2.$
 that not be-PST-3SG.INDF enough girl the party-IN
 'Yesterday I inferred that Mari left only from the fact that there were not enough girls at the party.'

- (85) [_{Foc} CSAK AR-RA ____₂] következtet-t-em₁ [_{PART} ab-ból ____₃] ____₁ tegnap [hogy Mari haza men-t-Ø]₃
 only it-EL infer-PST-1SG.DEF it-SUB yesterday that Mari home go-PST-3SG.INDF
 [hogy nem vol-t-Ø elég lány a buli-ban]₂.
 that not be-PST-3SG.INDF enough girl the party-IN
 ‘The only thing I inferred yesterday from the fact that Mari went home was that there were not enough girls at the party.’

Essentially, the clauses moving out from these small clauses to the right periphery show the mirror image of the order of the small clauses in the left-periphery as depicted in (86): the clause moved out from Spec,FocP follows the clause moved out from Spec,TP. I propose that (86) arises because of the constraint in (87): the movement of clauses must happen as soon as the violation of (83) arises. Thus, once the first small clause moves to Spec,TP, the embedded clause re-merges with the next projection, which is TP; once the second small clause moves to Spec,FocP, the embedded clause re-merges with the next projection, which is FocP. (Again, that the clauses are external to VP is shown by their position after *tegnap* ‘yesterday’.)

- (86) [_{FocP} [_{SC} az [hogy ...]]] [_{TP} [_{SC} az [hogy ...]]] ... [VP ...] ... ✓ TP] ✓ FocP]

- (87) If a clause cannot occur in Spec,βP, it must re-merge with βP in the next derivational step.

In sum, by (74) of the previous section the embedded clause must c-command its expletive at every derivational step, i.e. the embedded clause is at least as high as the expletive. By (87) and economy, I assume that unless other factors play a role (e.g. *size* in Section 5), the expletive’s left-periphery position indicates the phrase with which the embedded clause re-merges.

3.3 Position-dependent opacity

The previous subsections showed that Hungarian embedded clauses can occur in various left-periphery positions. Although moved or adjoined clauses have been handled as absolute islands in the earlier syntactic literature (Ross 1967; Wexler and Culicover 1983), the more recent literature has found that subextraction can proceed from re-merged clauses (e.g. Sauerland 1999; Collins 2005). Thus, building on the idea that moved constituents are *not* islands, this section describes how the movement and *position* of the embedded clause constrains subextraction possibilities. The generalization drawn in the section will be similar to a clause-specific version of the Generalized Proper Binding Condition (Fiengo 1977; Lasnik and Saito 1994), which claims that subextraction from a clause must land higher than the final position of the clause. In short, the movement of the clause cannot be remnant movement.

The terminological separation of the moving constituents makes the discussion easier to follow. The term *re-merge* will only refer to the movement of clauses (extracted from small-clauses), while the term *subextract* will only refer to movement from within clauses (i.e. particle-, focus-, and relative-movement). At this point, this is a descriptive distinction; how these refer to different operation-types will be discussed in Section 5.1. The terms *move* and *land* will be kept as general terms that can refer to both clauses and constituents smaller than clauses.

As Section 3.1 described, the only left-periphery positions in which embedded clauses or small clauses containing the embedded clauses can re-merge are the topic positions, which I defined as sisters to maximal FocPs or ComPs. Example (88) shows that even if a clause is transparent to the subextraction of particles in a post-verbal position, particle extraction is not compatible with the subsequent topicalization of the clause. That is, even if we do not expect the impossibility of particle movement on the basis of *clause-internal size* based on Section 2, the clause’s position can introduce blocking effects. Notice that it does not matter whether the clause is directly pre-verbal, i.e. it is sister to a maximal FocP, or it precedes a sentential adverb, i.e. it is sister to a maximal ComP. Furthermore, it does not matter whether the constituent undergoing movement is just the embedded clause or the small clause containing the expletive and the embedded clause.

- (88) [_{ComP/FocP} ... [(az-t) hogy ... XP] ... [TP × [... [VP V [hogy ... XP]]]]]
- ↓
- {_{TOP} * (az-t,) hogy ____₂ fus-s-ak a város-on)₁ (_{SADV} szerencsére) [_{PART} keresztül]₂ akar-t-a
 it-ACC that run-SUBJ-1SG.INDF the city-SUP fortunately across want-PST-3SG.DEF
 ____₁, { hogy ____₁ fus-s-ak a város-on)₁
 that run-SUBJ-1SG.INDF the city-SUP
 ‘(Fortunately,) she wanted me to run across the city.’

The example in (89) shows that focus subextraction is also incompatible with the subsequent topicalization of the clause. Again, it does not matter whether the clause precedes an adverb (i.e. whether it re-merges with FocP or CompP), and whether the clause re-merges on its own or as a part of the small-clause containing the expletive.

- (89) $[_{\text{Comp/FocP}} \dots [(az-t) \text{ hogy } \dots \text{XP}] \dots [_{\text{FocP}} \times [\dots [_{\text{VP}} \text{ V } [\text{hogy } \dots \text{XP}]]]]]$
- $\langle_{\text{TOP}}^* (az-t) \text{ hogy } [_{\text{PART}} \text{ keresztül}]_3 \text{ fut-ott-am} \dots_2 \dots_3 \rangle_1 (sajnos) [_{\text{Foc}} \text{ CSAK A VÁROS-ON}]_2$
 it-ACC that across run-PST-1SG.INDF fortunately only the city-SUP
hall-ott-a $\dots_1, \langle \text{hogy } [_{\text{PART}} \text{ keresztül}]_3 \text{ fut-ott-am} \dots_2 \dots_3 \rangle_1$
 hear-PST-3SG.DEF that across run-PST-1SG.INDF
 ‘She heard that I ran across ONLY THE CITY.’

However, topic clauses re-merged with matrix maximal FocP or CompP projections are not islands; relative pronouns can be subextracted from them as in (90). The embedded clause can precede or follow sentential adverbs i.e. it can be re-merge with FocP or CompP. Notice that the matrix-modifying sentential adverb showing up between the subextracted relative pronoun and the embedded clause of origin rules out clausal pied-piping as an option because it shows that the relative pronoun leaves the embedded clause. Finally, the subextraction is grammatical from the FocP/CompP-sister embedded clause just as it is from the FocP/CompP-sister small clause containing the expletive and the embedded clause.

- (90) $[_{\text{RelP}} \checkmark \dots [_{\text{Comp/FocP}} \dots [(azt) \text{ hogy } \dots \text{XP}] \dots [_{\text{FocP}} \dots [_{\text{VP}} \text{ V } [\text{hogy } \dots \text{XP}]]]]]$
- Ez az a város, [REL amelyik-en]₂ (sajnos) [TOP (az-t) hogy [PART keresztül]₃ fut-ott-am*
 this that the city which-SUP unfortunately it-ACC that across run-PST-1SG.INDF
 $\dots_2 \dots_3 \rangle_1 (sajnos) [_{\text{Foc}} \text{ CSAK MARI}] \text{ hall-ot-a} \dots_1$
 unfortunately only Mari hear-PST-3SG.DEF
 ‘This is the city which, that I ran across, only Mari heard it unfortunately.’
 (Context: I wanted it to be heard by more people)

In sum, clauses that get later topicalized (re-merged with CompP or FocP) are opaque to particle and focus subextraction to matrix Spec,TP and Spec,FocP, respectively but topicalization can be followed by relative extraction to Spec,RelP. The generalization in (91) summarizes these facts: A clausal projection αP (or the small clause containing it) cannot undergo (remnant) movement to a left-periphery position immediately dominated by βP if movement originating in αP lands in a lower left-periphery position Spec, γP (i.e. if $\beta \geq \gamma$). For example, since FocP/CompP-sister topic clauses (or topic small clauses) c-command Spec,FocP and Spec,TP, if focus or particle moves out from some clause, the clause cannot be topicalized. (Since (83) excludes any leftward-moved clause c-commanded by any focus, the Foc1P-Foc2P-distinction is irrelevant; every topic clause c-commands each foci.) In contrast, Spec,RelP c-commands the re-merger sites of topic clauses and small clauses, hence, (91) does not constrain relative movement from topic clauses and small clauses.²¹

- (91) **Position-dependent opacity:** αP or the small clause containing αP cannot move to a position immediately dominated by a βP , if some XP from αP moved to Spec, γP , and $\alpha, \beta, \gamma \in \text{fseq F}$, and $\beta \geq \gamma$.

$$[_{\beta\text{P}} \times \dots [_{\gamma\text{P}} \text{XP} [\dots [_{\text{VP}} \text{V } [_{\alpha\text{P}} \text{hogy } \dots \text{XP}]]]]]$$

²¹Anikó Lipták (p.c.) asks whether position-dependent opacity must be defined in terms of hierarchy, or whether the linear definition “clause of origin cannot precede subextracted material” would suffice. One possible argument against this linear formulation and in favour of my structural formulation comes from the movement of N(egative) C(oncord) I(tems). According to É. Kiss (2010), NCIs must adjoin to the NegP/PolP headed by the negative Neg. Embedded NCIs cross the clause boundary and adjoin to either to the right or left of the matrix NegP/PolP as in (i). If the clause is post-verbal, the rightward NCI-movement lands behind (and hierarchically above) the embedded clause, which is incompatible with the linear but compatible with the hierarchical formulation of position-dependent opacity.

- (i) *Mari* $\langle_{\text{POL}} 'senki-t\rangle_1 [_{\text{PolP}} \text{nem akar-t-a}_2 \dots_2, [\text{hogy } [_{\text{PART}} \text{meg}] \text{ puszil-j-}\emptyset \dots_1]] \langle_{\text{POL}} 'senki-t\rangle_1$
 Mari no.one-ACC not want-PST-3SG.DEF that part kiss-SUBJ-2SG.INDF no.one-ACC
 ‘Mari did not want you to kiss anyone’ lit. ‘Mari did not want you to kiss no one.’

The movement of NCIs is size-sensitive: Tensed clauses like (ii) are opaque to NCI-extraction. This fits the picture presented in Section 2 if $\text{fseq}=\dots\text{T}>\text{Pol}>\text{sMod}$.

- (ii) **Mari* $\langle_{\text{POL}} 'senki-t\rangle_1 [_{\text{PolP}} \text{nem } \{ \text{hall/hazud} \} \text{-t-a}_2 \dots_2, [\text{hogy } [_{\text{PART}} \text{meg}] \text{ puszil-t-tál} \dots_1]] \langle_{\text{POL}} 'senki-t\rangle_1$
 Mari no.one-ACC not hear/lie-PST-3SG.DEF that PART kiss-PST-2SG.INDF no.one-ACC
 Intended: ‘Mari did not {hear/ lie} that you kissed anyone.’

The generalization in (91) is basically a ban on remnant movement restricted to clausal projections. That is, it is a restricted version of the Generalized Proper Binding Condition (Fiengo 1977; Lasnik and Saito 1994), which claims that traces must be bound throughout a derivation. Importantly, (91) does not claim that all cases of remnant movement are ungrammatical. Section 5.2 discusses cases in which remnant movement of non-clausal projections is possible in Hungarian.

Another comparable generalization to (91) is the *Müller-Takano Generalization* (Müller 1993; Takano 1994), which says that the subextraction from some αP cannot be the same type as the remnant movement of αP . The movement of relative pronouns, topics, quantifiers, foci, and particles can be all categorized as $\bar{A}$ -movement (or movement landing in $\bar{A}$ -positions) in Hungarian. Müller-Takano-style reformulation of (91) would be the following: A clausal constituent with a left-periphery (i.e. topped some $\bar{A}$ -projection) cannot undergo $\bar{A}$ -remnant movement if $\bar{A}$ -subextraction has taken place from within it. In other words, the only way (91) is more restrictive than the Müller-Takano Generalization is that (91) also constrains the type of the remnant itself, not just the type of the movement operations. Again, this restriction will turn out to be crucial in Section 5.2 as nominal remnants in Hungarian can undergo $\bar{A}$ -movement after $\bar{A}$ -subextraction has taken place from them.

3.4 Expletive-mediated position-dependent opacity

While the positions in which the embedded clauses themselves can be pronounced are highly restricted, the small-clause remnant analysis of the previous sections open up more testing possibilities: The left-periphery position of the clausal expletive indicates the attachment height of embedded clauses in the right periphery (see Sections 3.1 and 3.2.2).

The generalization repeated in (92) (established in Section 3.1) has the immediate consequence that clausal expletives indicate the left-periphery position of the small clause from which the embedded clause undergoes movement. Furthermore, (93) (repeated; established in Section 3.2.2) adds that embedded clauses re-merge with the lowest possible phrase. In short, even if whole clauses are banned from a number of left-periphery positions, their expletives occurring in the left-periphery indicate their lowest re-merger-height in the right-periphery. Because of this, the rest of the section uses expletives in the matrix left-periphery to bring further examples for the position-dependent opacity generalization in (91).

(92) A clausal expletive must be bound by a c-commanding clause associated with it at each derivational step.

(93) If a clause cannot occur in Spec, βP , it must be re-merged with βP in the next derivational step.

The example in (94) indicates that the subextraction of particle *keresztül* ‘across’ is only possible if the small clause is VP-internal and the expletive is *pro*-dropped. (The embedded clause itself needs to leave the VP by (76).) Particle subextraction is incompatible with the subsequent movement of the small clause to either Spec,TP, Spec,FocP or re-merges with Comp or FocP (above or below sentential adverbs), as it is indicated by the banned expletive positions in (94). Thus, at least for particle movement, it holds that subextraction is not compatible with overt expletives (cf. Lipták 1998; É. Kiss 2002). (In the bracket diagrams to come, rightward pointing dashed arrows to another left-periphery position indicate that a clause whose expletive shows up in a higher position is earlier opaque to subextraction to a lower position.)

- (94) [ComP [SC *az* [*hogy* ... XP]] [FocP [SC *az* [*hogy* ... XP]] [TP [SC *az* [*hogy* ... XP]] × ✓ [... [VP V [SC *pro* [*hogy* ... XP]]]
- Mari $\langle_{TOP}$ **az-t* $_2$ $\rangle$ ($_{SADV}$ *szerencsére*) $\langle_{FOC}$ **CSAK* *AZ-T* $_2$ $\rangle$ $\langle_{PART}$ **az-t* $_2$ $\rangle$ [$_{PART}$ *keresztül*]₁
 Mari it-ACC fortunately only it-ACC it-ACC across
akar-t-a $\langle_{pro}$ $_2$ $\rangle$, [*hogy* *fus-s* *a város-on* $_1$]₂
 want-PST-3SG.DEF it.ACC that run-2SG.SUBJ.INDF the city-SUP
 VP-internal expletive: ‘Mari wanted you to run across the city.’

Focus movement reveals that overt expletives do not bring about absolute islandhood. As (95) shows, VP-internal *pro*-dropped expletives as well as overt expletives in the Spec,TP particle position are compatible with long focus-extraction from the clause associated with the expletives. At the same time, focus extraction cannot proceed from clauses whose small clause subsequently undergo topicalization, i.e. re-merges with some maximal FocP or ComP position as indicated by the expletive positions in (95).

- (95) [Comp [SC az [*hogy*...XP]] [FocP [SC az [*hogy*...XP]] [FocP × ✓ [TP [SC az [*hogy*...XP]] [... [VP V [SC pro [*hogy*...XP]]
Mari <_{TOP} **az-t* ___2> (<_{SADV} *szerencsére*) [_{FOC} *melyik város-on*]₁ *akar-t-a*₃ <(_{PART} *az-t* ___2) ___3
Mari it-ACC fortunately which city-SUP want-PST-3SG.DEF it-ACC
 <*pro* ___2>, [*hogy* [_{PART} *keresztül*]₄ *fus-s* ___1 ___4]₂?
 it-ACC that across run-SUBJ.2SG.INDF
 VP-internal or Spec,TP expletive: ‘Which city did you want them to run across?’

As introduced in Section 2.1, Hungarian has multiple focus projections: Foc2P dominating Foc1P, only distinguishable in clauses with multiple foci. In the a. version of (96), focus extraction to Spec,Foc2P can proceed from a clause whose small clause occurs in Spec,Foc1P, i.e. long focus movement lands in front of the expletive. This contrasts with the b. version of (96): Focus extraction to Spec,Foc1P is not possible from a clause whose small clause subsequently moves to Spec,Foc2P.

- (96) [Foc2P ✓ [Foc1P [SC az [*hogy*...XP]]]... BUT [Foc2P [SC az [*hogy*...XP]] [Foc1P × ...
 a. [_{FOC} CSAK EZ-EN A VÁROS-ON]₁ *akar-t-am*₃ [_{FOC} CSAK AZ-T ___2]₄ ___3 ___4, [*hogy* [_{PART} *that*
keresztül]₅ *fus-s* ___1 ___5]₂
 across run-SUBJ.2SG.INDF
 ‘The only city I wanted you only to run across was this one.’ (Context: Every other city, I wanted you to visit properly.)
 b. * [_{FOC} CSAK AZ-T ___2]₄ *akar-t-am*₃ [_{FOC} CSAK EZ-EN A VÁROS-ON]₁ ___3 ___4, [*hogy* [_{PART} *that*
keresztül]₅ *fus-s* ___1 ___5]₂
 across run-SUBJ.2SG.INDF

Finally, the expletive positions in (97) indicate that relative movement can proceed from clauses whose small clause ends up VP-internally, in Spec,TP, in Spec,Foc, or in a FocP/Comp-sister topic position.

- (97) [RelP ✓ [Comp [SC az [*hogy*...XP]]] [FocP [SC az [*hogy*...XP]] [FocP [TP [SC az [*hogy*...XP]] [... [VP V [SC pro [*hogy*...XP]]
Ez az a város, [_{REL} *amelyik-en*]₁ *Mari* <_{TOP} *az-t* ___2> (<_{SADV} *szerencsére*) [_{FOC} CSAK AZ-T ___2> *nem*
 this that the city which-SUP Mari it-ACC fortunately only it-ACC not
akar-t-a <(_{PART} *az-t* ___2) <*pro* ___2>, [*hogy* [_{PART} *keresztül*]₃ *fus-s* ___1 ___3]₂
 want-PST-3SG.DEF it-ACC it-ACC that across run-2SG.SUBJ.INDF
 VP-internal or Spec,TP expletive: ‘This is the city which Mari wanted you to run across.’
 Focused expletive: ‘This is the city which Mari ONLY wanted you to run across.’
 Topic expletive: ‘This is the city for which the following is true: That you run across it, Mari did not want that.’

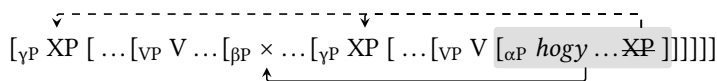
The table in (98) summarizes the position-dependent data so far. Assuming expletives indicate the position of small clauses, the general pattern is that subextraction can land in a position c-commanding the small clause but the small clause cannot end up in a position c-commanding the subextracted material. Particle extraction to Spec,TP is the most constrained as it can only proceed from clauses whose small clause remains VP-internal (and thus the expletive *pro*-dropped). Focus movement to Spec,FocP can only proceed from clauses whose small clause (expletive) is VP-internal, in Spec,TP, or in Spec,Foc1P (the focus subextraction in the last case lands in Spec,Foc2P). Relative movement to Spec,RelP can proceed from clauses whose small clauses is either VP-internal, in Spec,TP, in Spec,FocP, or is a FocP/Comp-sister (again, indicated by the expletive positions). Finally, notice that clauses occurring in a FocP/Comp-sister topic position exhibit the same opacity as clauses whose small clause occurs in the same FocP/Comp-sister position.

(98) **Opacity depending on the position of the expletive and the clause**

	Particle extraction to Spec,TP	Focus/ <i>wh</i> -extraction	Relative extraction to Spec,RelP
VP-internal expletive (<i>pro</i> -dropped)	✓	✓	✓
Expletive within Spec,TP (particle position)	×	✓	✓
Expletive within Spec,Foc1P (focus position)	×	✓ (to Spec,Foc2P)	✓
Expletive within Spec,Foc2P (focus position)	×	×	✓
Expletive within ComP/FocP-sister (topic position)	×	×	✓
ComP/FocP-sister clause (topic position)	×	×	✓

Under (92), the clause is *at least* at the height of the expletive. That is, the position of the expletive can be used to declare the *lowest clause-position we have evidence for*. (In the same way as overt left-periphery material in the left-periphery indicated *minimal* clause size in Section 2.3.) That is, the table in (98) can be re-interpreted in terms of clausal re-merger sites. Thus, the generalization in (99) gains further support from possible small clause positions, as indicated by the overt expletive positions.

- (99) **Position-dependent opacity:** αP or the small clause containing αP cannot move to a position immediately dominated by a βP , if some XP from αP moved to Spec, γP , and $\alpha, \beta, \gamma \in fseq F$, and $\beta \geq \gamma$.



So far, (99) has been only tested for biclausal structures: If a clause of origin and a subextracted constituent land in the same left-periphery, the subextracted constituent must c-command the clause. However, consider the tricausal structure in (100): Particle movement from the deepest clause cannot proceed to the matrix Spec,TP if the expletive associated with the deepest clause occurs in the Spec,FocP of the intermediate clause, i.e. if the deepest clause is adjoined to the intermediate FocP (or higher). That is, it is not only the case that movement from a clause in Spec,FocP cannot land in the Spec,TP of the same clause, but it cannot land in the Spec,TP of any clause (for a similar multi-clausal generalization, see Abels 2008). This general pattern is also depicted in the bracket diagram in (99).

- (100) [TP × [... V [... [FocP [SC *az* { *hogy* ... XP }] [TP...]]]]
 [_{PART} *Keresztül*]₁ *kell-ett* [*pro* ₃], [*hogy* <_{FOC} *CSAK *AZ-T* ₂> *akar-j-am* <*pro* ₂>, [
 across be.needed it that only it-ACC want-SUBJ-3SG.DEF it.ACC
hogy fus-s-anak a város-on ₁]₂]₃
 that run-3PL.SUBJ.INDF the city-SUP
 ‘It was necessary for me to want to run across the city.’

4 The limitations of previous accounts focusing on internal size

Beyond showing that Hungarian exhibits both size- and position-dependent opacity, the aim of this article is to provide a *unified account* for these two types of selective opacity. Previous accounts of the Williams Cycle and selective opacity in general derived the opacity of constituents from their *internal properties*. In other words, these accounts aimed to derive data exhibiting size-dependent opacity (i.e. data of the style seen in Section 2). Thus, in *delayed substitution* accounts, (Williams 2003; Poole 2023) it is the maximal projection of the embedded clause that determines when it becomes visible for matrix operations. In Abels’ (2012) account, it is the amount and type of left-periphery constituents in the embedded clause that determines which constituents within the embedded clause are accessible while obeying Relativized Minimality. In Keine’s (2016; 2019; 2020) account, it is the feature content of the maximal projection of the embedded clause that allows or blocks subextraction.

If one wants to keep the same mechanism and reasoning of these previous accounts while deriving position-dependent opacity (introduced in Section 3), the most natural option is to connect the position of a constituent to its internal structure: If a constituent wants to show up in a certain position, it must be of a certain size. However, I demonstrate in this section that this way of reducing position-dependent in (101) opacity to size-dependent opacity in (102) does not cover all cases because size and position do not always go hand in hand. After this, I exemplify the limitations of

accounts focusing on internal size with Keine's (2016; 2019; 2020) *horizon*-based account. Although space does not allow me to discuss other accounts, the argumentation can be extended to any of the above mentioned frameworks.²²

- (101) **Position-dependent opacity:** αP or the small clause containing αP cannot move to a position immediately dominated by a βP , if some XP from αP moved to $\text{Spec}, \gamma P$, and $\alpha, \beta, \gamma \in fseq$ F , and $\beta \geq \gamma$.
- (102) **Size-dependent opacity/Strict Williams Cycle:** Movement to $\text{Spec}, \beta P$ cannot proceed across a dominating αP , where $\alpha \geq \beta$ in the functional sequence.

Cable (2007; 2010b; 2010a) claims that *wh*-phrases do not belong to various categories (DPs, PPs, AdvPs etc.) but *wh*-movement uniformly targets QPs. In other words, the *position* of *wh*-words in Spec, CP is connected to their *size* because all constituents landing in a certain $\bar{A}$ -position must be of a certain size/category. This intuition of connecting size and position is generalized in (103), which claims that an embedded clause occurring in $\text{Spec}, \beta P$ must be a βP or larger, i.e. embedded clauses occurring higher in the matrix are larger. For example, if an embedded clause is contained in matrix $\text{Spec}, \text{Com}P$, this embedded clause must be at least a $\text{Com}P'$. By (102), movement to $\text{Spec}, \text{Foc}P$ is blocked from this $\text{Com}P$ clause because $\text{Com} > \text{Foc}$ in *fseq*. On the other hand, if the clausal expletive occurs in Spec, TP , the clause itself is possibly a TP , from which focus extraction can proceed to $\text{Spec}, \text{Foc}P$ because $\text{Foc} > T$. Thus, in theory, position-dependent opacity in (101) is redundant if it can be traced back to the combination of size-dependent opacity in (102) and the position-size correlation in (103).

- (103) **The position-size correlation:** If a clause is contained in $\text{Spec}, \beta P$, this clause must be αP such that $\alpha \geq \beta$ in *fseq*.

The size-position correlation in (103) predicts that position- and size-based opacity always go hand in hand. If a constituent originating in the embedded clause can move to the matrix position X across another constituent in the Y position of the embedded clause, movement to matrix X should not be blocked if the embedded clause occurs in the Y position of the matrix clause. For example, Hungarian particles occurring in the embedded clause's left-periphery do not block long *wh*- and focus movement as in (39), and *wh*- and focus movement is possible from clauses whose expletive occurs in Spec, TP as in (95). So far, topics and quantifiers have been left out from the discussion. Egressy (to appear) describes that neither topics nor quantifiers in the embedded clause block the long movement of particles, foci, and relative pronouns (partially also described by Koopman and Szabolcsi 2000). In (22), the universal quantifier *holnap* is 'also tomorrow' headed by the additive particle *is* 'also' does not block the long movement of the focus *csak a város* 'only the city' even if quantifiers can precede foci and sentential adverbs within clauses.

- (104) $\downarrow$ -----
[$\text{Foc}P$ ✓ ... [VP ... [quantifier *is* [... XP ...]]]]

[Foc CSAK A VÁROS-ON]₁ akar-t-am₂ _____, [*hogy* [_Q *holnap* is] keresztül₃ fus-s-anak _____]
only the city-SUP want-PST-1SG.INDF that tomorrow also across run-SUBJ-3PL.INDF
'I wanted them to run across ONLY THE CITY also tomorrow.'

If the quantifier *holnap* is 'also tomorrow' in (104) adjoins to $\text{Foc}1P$ and gets dominated by another $\text{Foc}1P$ (i.e. not a $\text{QP}/\text{Dist}P$ dedicated to quantifiers as in Brody and Szabolcsi 2003), it is in line with the size-dependent opacity in (102) that this quantifier does not block focus subextraction to matrix $\text{Spec}, \text{Foc}2P$. If there are no dedicated quantifier-projections in the clausal *fseq*, clauses whose expletive occurs in a quantifier position should not have any special opacity-reduction according to (103). This prediction is not borne out: If the expletive associated with an embedded clause combined with the additive *is* occurs in a preverbal quantifier position, the subextraction of foci is blocked from it:

²²Müller (2014a; 2014b) proposes a framework that determines the grammaticality of movement operations on the basis of movement paths arising from the crossed maximal projections. Since the movement path of cross-linguistic movement operations has a section in the matrix clause, this account considers clause positions. Although my proposal in Section 5 is similar to Müller's in certain respects, Müller only considers sub-extraction clauses that are complements to V . Furthermore, my analysis in Section 5.2 provides a principled explanation why domains that can be escaped successive cyclicly do not exhibit size- and position-dependent opacity. Since Müller's analysis of size-dependent opacity relies on the idea that movement must stop in every phrase, his analysis is not able to connect successive cyclicity and the lack of size-dependent opacity.

- (105) $[_{\text{FocP}} [\text{SC azt is } \boxed{\text{hogy} \dots \text{XP}}]]_{\text{FocP}} \times \dots$
 $[_{\text{Q}} \text{Az-t is } __4]_1 \langle_{\text{FOC}}?? \text{CSAK A VÁROS-ON}\rangle_2 \text{akar-t-am}_3 __3 __1,$
 it-ACC also only the city-SUP want-PST-1SG.DEF
 $[\text{hogy } \langle_{\text{FOC}} \text{CSAK A VÁROS-ON}\rangle_2 \text{fus-s}_5 [_{\text{PART}} \text{keresztül}]_6 __5 __6 __1]_4$
 that only the city run-2SG.SUBJ.INDF across
 With the embedded focus: ‘I also wanted you to run across ONLY THE CITY.’ (I want multiple things.)

The same problem occurs with topics. Example (106) indicates that topics do not block focus extraction from embedded clauses. As long as topics do not have a dedicated RefP/TopP projection (as in Brody and Szabolcsi 2003) but are able re-merge with maximal Foc1P projections, the grammaticality of focus to Spec,Foc2P extraction is compatible with (103). However, topic-clauses and clauses whose small clauses are topics do exhibit position-dependent opacity as the examples of Sections 3.2 and 3.4 indicated. The impossibility of focus extraction would be expected under (103) only if topic clauses were TopPs/RefPs such that Top/Ref > Foc in the *fseq*.

- (106) $[_{\text{FocP}} \checkmark \dots [_{\text{VP}} \dots [\text{topic } [\dots \text{XP} \dots]]]]$
 $[_{\text{FOC}} \text{CSAK A VÁROS-ON}]_1 \text{akar-t-am}_2 __2, [\text{hogy } [_{\text{TOP}} \text{holnap}] \text{keresztül}_3 \text{fus-s-anak } __1 __3]$
 only the city-SUP want-PST-1SG.INDF that tomorrow across run-SUBJ-3PL.INDF
 ‘As for tomorrow, I wanted them to run across ONLY THE CITY.’

In sum, the attempt to reduce position-dependent opacity in (101) to a combination position-dependent opacity (102) and the position-size correlation in (103) is unsuccessful. Such reduction misses the point that clauses in topic and quantifier positions exhibit position-dependent opacity even if the presence of topics and quantifiers in the embedded left-periphery do not affect the size-dependent opacity of clauses. In contrast, notice that the generalization on position-dependent opacity in (101) captures this nuance by defining the banned position of α P (or the small clause containing α P) as a position immediately dominated by β P, which does not have to be a dedicated specifier of β P but can be an adjunct, too. For this reason, my proposal in Section 5 basically turns around (103) by claiming that larger clauses must be in a higher position, i.e. size implies position rather than the way around.

One of the most recent accounts of selective opacity is provided by Keine (2016; 2019; 2020). The core of this proposal is that probes responsible for various syntactic operations have various *horizons*, which specify when their search gets terminated, i.e. which projections serve as barriers for probes. For example, if a probe on T has a horizon C, then subject raising from CPs is not possible because T’s search is terminated by the C feature of CP.

In order to derive size-dependent opacity in (102), one can propose the following entailment between a probe’s height and its horizons:²³

- (107) If a probe is located on β , it has α as its horizon if $\alpha \geq \beta$ in *fseq*.

For example, the ban on particle movement from FocPs could be derived as follows: Since the probe associated with particle movement is on matrix T, its search is terminated by the Foc feature of an embedded FocP because Foc > T in *fseq*. In a similar way, since the probe associated with focus movement is on matrix Foc, its search is terminated by the Com feature of an embedded ComP because Com > Foc in *fseq*.

Under the *horizon*-based account, whether a constituent is opaque to an operation depends only on its internal structure, more specifically, the features of the internal projections of the constituent. One attempt to extend the account to position-dependent opacity in (101) could involve the position-size correlation introduced in (103). For example, since a focused clause needs to be a FocP at least, the focused clause has the same opacity as a clause with a focus; in both cases, particle probes on T are terminated by the focus feature because Foc > T. However, as discussed above, (103) cannot deal with topics and quantifier because topics and quantifiers do not increase the opacity of embedded clauses but embedded clauses occurring as topics and quantifiers have a higher degree of opacity. More specifically, the *horizon*-based account faces the problem that the probe responsible for focus movement should be terminated by

²³Keine’s original formulation is shown in (i). Notice that (i) is less restrictive than (107): while (107) derives the strict Williams Cycle, (i) only excludes lower horizons for higher probes, e.g. no probe with the horizon T can be on Foc or Rel.

(i) **Locality $\rightarrow$ height entailment:** Given an extended projection $\Phi = \langle \Pi_n > \Pi_{n-1} > \dots > \Pi_1 \rangle$, for any nonvacuous prope π , $[\dots]$ if π has Π_m as a horizon, then π cannot be located on the projection $\{\Pi_n, \dots, \Pi_{m+1}\}$ (Keine 2019:p. 50)

some topic/quantifier feature if the embedded clause occurs as a topic/quantifier but not if the embedded clause has a topic/quantifier.

Another problem arising with (103) is that Hungarian lacks remnant movement only in the clausal domain. As it will be exemplified by DP remnant movement in Section 5.2, there are projections in Hungarian that can undergo remnant movement in the left-periphery (for the remnant movement of *v*Ps and PP, see Koopman and Szabolcsi 2000; Dékány and Hegedűs 2015). That is, if we want to stick to a constraint like (103), we need to restrict it in a rather *ad hoc* way: a clause occurring in Spec,βP must be βP, but this does not hold to other projections, e.g. focused nominals. For example, focused clauses should be FocPs from which particle extraction is blocked, but focused nominals should not be FocPs, so particle subextraction should be possible from them. In a *horizon*-based derivation, (107) claims that focused clauses should have a Foc feature that is a horizon for the particle probe on T because Foc>T in *fseq*. At the same time, focused nominals should lack the Foc feature if they allow remnant movement to Spec,FocP after a particle has been subextracted from them to Spec,TP.

5 Proposal

As argued in Section 4, the opacity of Hungarian embedded clauses cannot be reduced solely to their *internal structural size*, that is, accounts focusing on size-dependent opacity are insufficient. In another family of accounts, the transparency of embedded clauses is influenced by their *syntactic environment* in the matrix clause. For example, Rackowski and Richards (2005), Van Urk and Richards (2015), and Den Dikken (2018) claim that the embedded clause becomes transparent if the matrix verb agrees with it. However, since these accounts operate with absolute terms of phasality (i.e. it is always the *v* that agrees with CP), they are unable to capture opacity-differences arising from internal size.

The core of my unified account for Hungarian size- and position-dependent opacity is that Hungarian embedded clauses cannot be escaped via successive cyclic movement, rather the clauses must re-merge with the matrix clause in order to be transparent or "unlocked" to cross-clausal operations. Size- and position-dependent opacity will be argued to follow from the properties and constraints of a single mechanism: the re-merger of embedded clauses. Unlike the previous accounts of Agree-based unlocking, the present account based on re-merger-based unlocking is size-dependent as larger embedded clauses need to re-merge higher in the matrix clause, that is, larger embedded clauses are unlocked later in the derivation. Furthermore, since re-merger brings transparency at the cost of mobility, re-merger of clauses and subextraction from them are only possible after the clause has reached its final position in the matrix clause. In short, the novelty of the re-merger-based account proposed here is that it considers *both* internal size *and* external environment.

Section 5.1 introduces the constraint *Move Along Increasing Functional Sequence Indices* (MAIF), which serves as a size-dependent alternative to the Phase Impenetrability Condition (PIC) (Chomsky 2000, 2001). In Section 5.2, I show that successive cyclic movement remains a canonical option to escape locality domains, but if a domain can be escaped in this way, it does not show size- and position-dependent opacity. In Section 5.3, I propose that the Hungarian left-periphery lacks the left-periphery specifiers that could be used as escape hatches for successive cyclic movement. Section 5.4 introduces the idea that Hungarian embedded clauses can be still escaped after they re-merge with an equal-sized matrix position. Section 5.5 derives size-dependent opacity because larger clauses re-merge higher, i.e. they become transparent later in the derivation. Finally, Section 5.6 and 5.7 argues that position-dependent opacity is derived because embedded clauses and the small clauses containing them are mobile only below the re-merger site, i.e. before becoming transparent to subextraction.

5.1 Move Along Increasing *fseq*-indices (MAIF)

Syntactic operations have been found to be constrained by locality domains, which have been called *phases* in the last decades. In standard phase theory (Chomsky 2000, 2001), the Phase-Impenetrability Condition (PIC) in (108) specifies the positions which are available from outside a phase. In Chomsky's original work, phasehood is *absolute* in the sense that it is rigidly associated with specific phrase labels. For example, the phrase headed by C is a phase and always a phase. Therefore, whenever the building of a CP is complete, no material c-commanded by the C head (i.e. nothing in the *domain* of C) can leave the CP. Further categories associated with phasehood including *v* and D (e.g. Chomsky 2000; Svenonius 2004) are handled identically. Fundamentally, the (un)availability of the material c-commanded by the phase-head is not dependent on the landing site of the movement relative to the size and position of the phase. In short, default phase theory relying on the PIC in (108) and phasehood rigidly associated with particular categories is *size- and position-insensitive*.

- (108) **Phase-Impenetrability Condition (PIC):** The domain of H is not accessible to operations outside HP; only H and its edge are accessible to such operations. (Chomsky 2000:p.13).



In contrast, Sections 2 and 3 argued that the opacity of the various clauses in Hungarian is sensitive both to *size* (e.g. a FocP is transparent to movement to Spec,RelP but not to movement to Spec,TP) and *position* (e.g. even if a clause is transparent to movement to Spec,FocP, it becomes opaque to this movement type if it gets topicalized later in the derivation). Therefore, the *absolute* definitions of standard phasehood insensitive to size and position are not sufficient while analyzing *selective* opacity in Hungarian.

To capture the facts of selective opacity in Hungarian, I propose an alternative to the PIC in (109). The constraint *Move Along Increasing fseq-indices* (MAIF) says that the launching position of a moving XP and its landing site must belong to the same extended projection. (Extended projections are tokens of functional sequences.) According to Chomsky (1977: p. 85), only clause-internal movement is universal; cross-clausal movement arises independently, via language-specific rules. MAIF is the generalization of this intuition: Syntactic movement universally proceeds within extended projections. The technical implementation of this idea is the following: Along the path of the movement, it must hold that the crossed heads belong to the same *fseq* and that their *fseq*-indices increase.²⁴

- (109) **Move Along Increasing *fseq*-indices (MAIF):** XP can move from A to B iff the heads crossed form a well-formed functional sequence fragment.

The term *crossing* is defined as follows:

- (110) **Crossing:** For any two heads α and β both c-commanding the original position of XP and both c-commanded by the landing site of XP, α is higher than β in the sequence of crossed heads by XP iff α asymmetrically c-commands β .

The term *well-formed functional sequence fragment* is defined in (111). In Grimshaw's 2000 terms, " α higher in *fseq* β " means " α has a higher *fseq*-index than β ". Thus, if *fseq* is $V < v < T < C$, the sequences $V-v-T$ and $v-T-C$ are well-formed *fseq*-fragments but $*V-v-C$ and $*C-V-v-T$ are not well-formed because of the missing T head and the "index-drop" between C and V, respectively. In order to make the sequences of crossed heads comparable to *fseqs*, I will unconventionally write *fseqs* starting with the lower heads. The points of crash will be marked with boldface.

- (111) **Well-formed functional sequence fragment:** It holds for every α - β two-head sub-sequence of the sequence of heads crossed by XP that
- (i) α and β belong to the same functional sequence,
 - (ii) α is higher than β in this functional sequence, and
 - (iii) there is no γ such that $\alpha > \gamma > \beta$ in this functional sequence.

The output of MAIF is similar to the spirit of Bošković (2014): Locality-boundaries coincide with the boundaries between extended projections.²⁵ In (112), CP' is a locality-boundary: XP cannot be subextracted from CP' in one step to a position c-commanding the matrix V because this movement step would proceed along the $*V'-v'-T'-C'-V$ sequence, which would violate (109) as $V < C$ in *fseq*. At the same time, TP' will not be a domain-boundary if it is selected by a C' , which is part of the same *fseq* as in (112): XP can move along the $V'-v'-T'-C'$ sequence to Spec, CP' . Nevertheless, the embedded TP' becomes locality-boundary in (113) once it is directly selected by a matrix V, which starts another *fseq*. In this case, an XP subextracted from TP' would have to move along the $*V'-v'-T'-V$ sequence.

- (112) $[\times [_{VP} V [_{CP'} C' [_{TP'} T' [_{vP'} v' [_{VP'} V' XP]]]]]]$ (113) $[\times [_{VP} V [_{TP'} T' [_{vP'} v' [_{VP'} V' XP]]]]]$

²⁴The idea of relating movement to a well-formed paths goes back at least to Kayne (1984); constraining movement paths in terms of category labels in *fseq* is reminiscent of the proposal in Müller (2014a; 2014b).

²⁵Bošković (2014) assumes that the clausal spine contains two *fseqs*, a *v-fseq* and a *C-fseq*. For now, I assume that clausal spines are a single *fseq*, and I discuss the separation of the *v-fseq* in Footnote 33.

However, while Bošković (2014) does not depart from the original PIC in (108) in that he keeps the concept of phases in the theory, MAIF in (109) does not make reference to any locality domains. Thus, for the purposes of syntactic movement under MAIF, the concept of phases could be dropped or loosened (e.g. to *Read only* suggested by Agarwal (2022; to appearb; to appeara)), although generally abandoning phases would raise questions on phase-related operations not necessarily involving movement e.g. ellipsis, Case, and Agree. The alternative is to build on Bošković’s (2014) idea and claim that extended projections are phases for the purposes of all operations. Under this option, MAIF is merely a technical implementation for movement. As far as this movement-oriented paper is concerned, there is no theoretical motivation to choose one of these options over the other. For practical reasons, I refer to the highest specifiers of extended projections as *edges*.

5.2 Successive cyclicity as a way around size- and position-dependent opacity

Following the intuition of Chomsky (1977: p. 85), I propose that individual languages use various strategies to make cross-clausal movement grammatical. If a language cannot make use of any strategy of that kind, cross-clausal movement is unavailable as it is the case in Georgian (Harris 1981; Bondarenko 2024) or Tsez (Polinsky and Potsdam 2001). In MAIF-terminology, cross-clausal movement must be turned into movement steps which are internal to clauses or extended projections. This subsection discusses *successive cyclicity* as one such strategy, Section 5.4 discusses clausal re-merger as an alternative strategy for MAIF-complaint long movement.

Successive cyclic movement is one way to avoid locality violations either under the head-based MAIF definition in (109) or the phrase-based PIC definition in (108). Essentially, both PIC and MAIF claim that within some locality domain αP , the movement of constituents c-commanded by α (in Chomsky’s 2001 terms, α ’s complement is spelled out) is banned whereas constituents asymmetrically c-commanding α (α ’s specifier(s)) are accessible from outside αP . In other words, the *edge* status of specifier(s) of locality domains is preserved under MAIF in (109): XP in the Spec, αP of some locality domain αP c-commands every head in α , nothing in αP constrains further movement of XP.²⁶ For example, while subextraction is impossible in one step from CPs and TP_s in (112) and (113), respectively, successive cyclicity makes cross-clausal movement MAIF-compliant. Assuming that *fseq* is $V < v < T < C$ for (114), if XP is complement to the embedded V' , XP can move to the embedded Spec,CP, as it moves along the $V'-v'-T'-C'$ sequence in the first movement step in (114). In the second step, the first head crossed is the matrix V, after which it only crosses matrix heads. In the same way, subextraction from TP_s can proceed in two steps via Spec,TP_s in (115): The first step proceeds along the $V'-v'-T'$ sequence and the matrix V is the first head to cross in the second step.²⁷

- (114) [✓ [VP V [CP' XP [C' [TP' T' [vP' v' [VP' V' XP]]]]]]] (115) [✓ [VP V [TP' XP [T' [vP' v' [VP' V' XP]]]]]]

In the following subsections, the core of my proposal will be that the size- and position-dependent opacity of Hungarian clausal domains arises because MAIF-violations cannot be avoided via successive cyclic movement in the absence of the necessary edge positions in the clausal domain. In contrast, if cross-clausal movement *can* proceed via clause-edges in some language, MAIF on its own does not pose any restrictions on the landing height. That is, the MAIF-based account predicts that if no independent factors constrain movement from a clause, size- and position-dependent opacity do not arise under successive cyclicity. Fong (2019) argues that successive cyclicity enables Mongolian to exhibit *hyperraising*, i.e. violations of the Williams Cycle or Size-dependent opacity repeated in (116) (for more discussion, see Halpert and Zeijlstra 2024).

- (116) **Size-dependent opacity/Strict Williams Cycle:** Movement to Spec, βP cannot proceed across a dominating αP , where $\alpha \geq \beta$ in the functional sequence.

²⁶The MAIF definition in (109) can be re-phrased in terms of dominating phrases: The category-labels in the sequence of phrases that dominate the launch site A but not the landing site B of XP must form a well-formed *fseq*-fragment. In the main text, I stick to the simpler head-based definition because the phrase-based definition requires a distinction between "dominated by every segment of" vs. "dominated by some segments of" (as in e.g. Chomsky 1986, 1995) to exclude αP from the set of crossed dominating phrases for movement launching in Spec, αP (i.e. to preserve the *edge* status of Spec, αP).

²⁷The framework allows extended projections to have zero, one, or infinitely many edges. For example, quantifier floating indicates that the Spec,TP position is an edge position for English subject raising: *The students_i seemed [TP' _i all win the race.]*. As an anonymous reviewer mentions, *wh*-extraction can take place from TP_s with a subject in Spec,TP: *Who_i did you expect [TP' _i me to invite _i]?* By default, nothing in the theory excludes the possibility of a second Spec,TP_s position in English, that serves as an escape hatch for the *wh*-movement. If the TP_s does not have a second specifier, it is still possible that English TP_s use the re-merger strategy presented in Section 5.4.

(117) [_{VP} ✓ [_v [_{VP} V [_{CP}' XP [_C' [_{TP}' T' [_{VP}' v' [_{VP}' V' [...XP ...]]]]]]]]]]
Bat < *nokhoi-g* *chang-aar* [_{CP} < *nokhoi-g*) *gailkhal-tai* *gej*] *khel-sen*.
 Bat dog-ACC loud-INSTR dog-ACC wonder-with COMP say-PST
 'Bat said loudly that dogs are wonderful.'

(118) **Position-dependent opacity:** α P or the small clause containing α P cannot move to a position immediately dominated by a β P, if some XP from α P moved to Spec, γ P, and $\alpha, \beta, \gamma \in fseq$ F, and $\beta \geq \gamma$.

(119) [_{TOP} *Tegnap*] *olvast-t-am* { [_{DP} [_{IP} *Mari level-ét*]] / [_{DP} *Mari-nak a* [_{IP} *level-ét*]] }
yesterday read-PST-1SG.DEF Mari letter-3SG.POSS-ACC mari-DAT the letter-3SG.POSS-ACC
‘Yesterday I read Mari’s letter.’

(120) [✓ [FocP ✓ × ... [VP V [DP possessor_{DAT} [D [IP possessor_{NOM} possessee]]]]]]

⟨DP —₁ A [IP *level-é-t*⟩ [TOP *tegnap*] [FOC CSAK Mari*(-nak)]₁ *olvas-t-am* ⟨DP —₁ a

the letter-3SG.POSS-ACC yesterday only Mari-DAT read-PST-1SG.DEF the

[IP *level-é-t*]

letter-3SG.POSS-ACC

‘Yesterday, I read ONLY MARI’s letter’

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In sum, if an extended projection has an edge, successive cyclic movement can proceed from within this projection. MAIF predicts that movement from an edge position can land on any height in the next highest extended projection without being constrained by (116) and the remnant of the extended projection can undergo movement without being constrained by (118).²⁹

5.3 Hungarian *hogy*-clauses lack edges

In this section, I propose that, while other languages and non-clausal domains in Hungarian make use of the *edge* positions for MAIF-compliant cross-clausal movement, the Hungarian clausal left-periphery does not have edge positions, i.e. successive cyclic movement is impossible. The specifiers introduced for Hungarian are Spec,RelP for relative pronouns, Spec,ComP for sentential adverbs, Spec,Foc1P and Spec,Foc2P for foci, and Spec, TP for particles. Following Rizzi's analysis for Italian (Rizzi 1997), I assume that these left-periphery positions are criterial positions and movement to these positions bring about *criterial freezing*: Constituents landing in these specifiers cannot move further, that is, these specifiers cannot be used as escape hatches. I assume, moreover, that sModP does not have a specifier.

That Hungarian left-periphery specifiers cannot be used as escape hatches is supported by the absence of visible reflexes via these positions. For example, Hungarian focus movement triggers a reflex at its landing site: As shown in Section 2.1, whenever Spec, FocP is filled, the verb moves in front of its particle in Spec,TP to the Foc head. Therefore, we would expect verb-particle inversion in embedded clauses if long-moving constituents used the embedded Spec,FocP as an escape hatch. However, this is not the case: Under long focus movement in (121), verb-particle inversion occurs only in the matrix clause with *el*, whereas the particle *keresztül* in the embedded clause stays in front of its verb (Horvath 1981, 1986; Richards 1997).

- (121) [FocP ✓ Foc+✓ [TP particle ... [VP V [Foc+× [TP particle ... [VP V [...XP ...]]]]]]
- [_{Foc} CSAK A VÁROS-ON]₂ ér-t-em₁ [_{Part} el] —₁ pro, [_{hogy} [_{Part} keresztül]₃
 only the city-SUP achieve-PST-1SG.INDF PART pro.ACC that across
fus-s-anak —₁ —₃]
 run-SUBJ-3PL.INDF
 'I accomplished that they would run across the ONLY THE CITY.'

The verb-particle inversion is also absent in the embedded clause of (57) and (56) suggesting that long relative movement does not have an intermediate stop in Spec,FocP', either.

The idea that Hungarian movement crossing clause-boundaries does not proceed via the edges of clauses is further supported by the fact that multiple operators may vacate a given clauses. In (122), the movement of the relative pronoun and the particle can proceed from the same clause. If the embedded clause had a single edge position, the successive cyclic movement of two operators would be impossible. Notice that the particle and the relative pronoun are separated in the matrix clause, so it is not reasonable to assume that these two operators cluster in the embedded clause and leave as a single constituent (as suggested by Grewendorf 2001). (In Section 2.3, the intermediate clause in (47) was also escaped by multiple operators.)

²⁹Due to space limitations, the main text cannot discuss two further edge-configurations. Firstly, apart from analyses arguing for *wh*-cluster-formation (Grewendorf 2001), the Bulgarian CP has been argued to have multiple specifiers/edges (Richards 1997, 2004; Zompì 2024). Furthermore, it has been found that Bulgarian has remnant-movement configurations like (i), where the subextracted *wh*-phrase ends up in the embedded Spec,CP and the *wh*-remnant containing it in the matrix Spec,CP (Richards 2004:p.459). (i) can get a MAIF-compliant derivation: The outer *wh*-phrase YP first moves to the embedded edge position Spec,CP, which is within-clausal movement in the embedded clause. After this, the inner *wh*-phrase XP can *tuck-in* to a lower Spec,CP in a MAIF compliant way as it only crosses heads within YP, i.e. this tucking-in step is MAIF-compliant. This can be followed by the movement of the remnant YP to the matrix Spec,CP. In sum, YP does not need to have an edge if CP has multiple edges: successive cyclic movement from within a YP in a Spec,CP is MAIF-compliant to another Spec,CP.

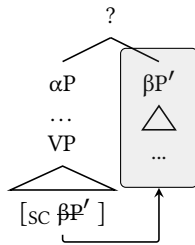
- (i) [✓ [VP V [_{CP'} [YP ... XP ...] XP [C' [TP' T' [_{VP'} v' [_{VP'} V' YP]]]]]]]
- [_{YP} Kolko studenti —₁]₂ se opitvaš da razereš [_{CP} —₂ [_{XP} ot koi strani]₁ e ubil Ivan —₂?
 how.many students REFL you.try to understand from which countries AUX killed Ivan
 '[YP how many students —₁]₂ are you trying to understand [CP —₂ [XP from which countries]₁ Ivan killed —₂?'

Secondly, an anonymous reviewer points out that MAIF does not exclude the movement of an extended projection with an edge to the edge of another extended projection. Such pattern is attested in colloquial English possessor extraction, e.g. *Who₁ did you say [_{CP} [_{DP} —₁ 's cake]₂ they really liked —₂]*? (Davis 2021:p.327). Since the possessor is at the edge of the larger DP moving to the edge of CP, the subextracted possessor does not cross any heads in the larger DP or the embedded CP. Thus, the possessor only crosses matrix heads while moving to matrix Spec,CP, which is MAIF-compliant. What motivates the pied-piping of the larger DP to Spec,CP is set aside here.

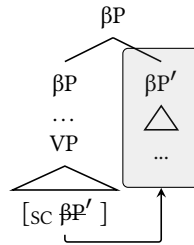
of the embedded clause and its adjunction site influence extraction possibilities, in particular, how symmetrical (as opposed to asymmetrical) re-merger facilitates subextraction. That is, my proposal is different from the proposal of Endo and Haegeman (2019), which assumes that embedded clauses always merge with equal-sized matrix projections. My proposal is more similar to that of Davison (2009), who claims that the re-merger of clauses may be both symmetric and asymmetric but symmetric structures exhibit different properties than asymmetric ones.

Assuming the embedded clause can re-merge with any matrix projection, the embedded clause's size may relate in three ways to the size of the matrix projection with which it re-merges. Firstly, the matrix projection can be larger than the embedded clause. This configuration is depicted in (126): If $\beta < \alpha$ in *fseq*, the re-merger of an embedded βP with a matrix αP yields a *smaller-to-larger* configuration. Secondly, the category label of the maximal projection of the embedded clause may be the same as the category label of the matrix projection with which it re-merges. In the symmetric configuration in (127), the embedded clause re-merges with an *equal-sized* matrix projection. Finally, the embedded clause can be larger than the matrix projection with which it re-merges. This configuration is depicted in (128): If $\beta < \alpha$ in *fseq*, the re-merger of an embedded αP with a matrix βP corresponds to a *larger-to-smaller* configuration.

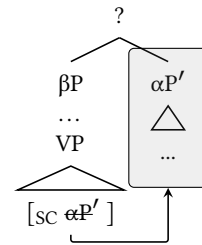
(126) *Smaller-to-larger*



(127) *Equal-sized*



(128) *Larger-to-smaller*

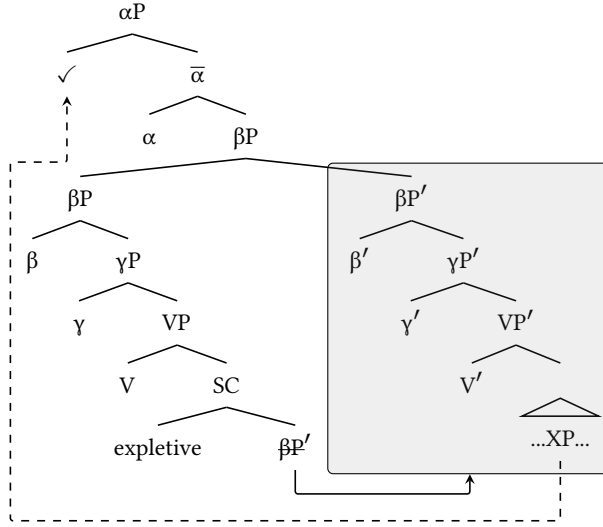


Since the *equal-sized* setting in (127) is symmetric, both βP s project. That is, the node immediately dominating the re-merger site takes the shared label of its children (as in Chomsky 2013), and becomes a shared maximal βP projection of the two lower projections. The labels of the nodes immediately dominating non-equal-sized matrix and embedded projections in (126) and (128) are not central to this paper, so I will label the nodes above these asymmetrical structures as "?".³¹

Size- and position-dependent opacity will be derived from the fact that re-merger opens the floor for MAIF-compliant cross-clausal movement only if the embedded clause re-merges with an *equal-sized* matrix projection. This is exemplified by the βP - $\beta P'$ re-merger in (129). Supposing the structure-building continues in the matrix clause even after the re-merger of the embedded clause as in (129), the sequence of heads c-commanding the XP in the re-merged embedded clause is the following: $V' - \gamma' - \beta' - \alpha$. Thus, under *fseq* = $V < \gamma < \beta < \alpha$, XP can move to Spec, αP because the *fseq*-indices of the heads keep increasing between the two copies of XP. In other words, the subextraction of XP to Spec, αP is compatible with MAIF in (125) because the embedded *fseq* finds a continuation in the matrix *fseq*. In other words, this apparent case of cross-clausal movement remains within a single extended projection because the highest projection of the embedded clause ceases to be the maximal projection in the extended projection after the re-merger of the embedded clause.

³¹One solution is to assume that in asymmetric configurations, the node above the re-merger site gets the label of the matrix projection. Another solution could be to assume with Starke (2001) and Adger (2003) that when two projections with category labels from the same *fseq* merge, the one with the higher *fseq*-index projects.

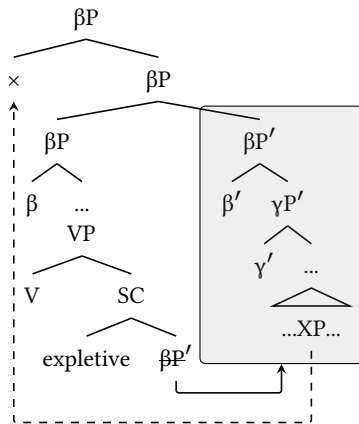
(129) **The embedded clause finds a continuation in the matrix clause**



Notice that as long as the embedded clause is VP-internal, the next highest head above β' is V, that is, β' tops a sequence of heads constituting a well-formed *fseq*-fragment. Thus, under MAIF in (125), the VP-internal $\beta P'$ is a barrier for movement: No XP can leave it as that movement operation would cross β' and then V in one step, which does not follow the increase-in-*fseq*-indices requirement. However, after the embedded clause has re-merged, the next highest head above β' is α , which belongs to the same *fseq* and has a higher *fseq*-index. Bošković (2014) suggests something similar: if a projection ceases to be the maximal projection in an extended projection, it ceases to be a phase.

Movement operations can only land in specifiers that c-command the re-merger site. Thus, subextraction to Spec, αP is possible in (129) but subextraction to Spec, γP is impossible. Moreover, within a certain projection, optional re-merger is generally assumed to happen after obligatory feature-checking movement operations (cf. e.g. Hunter 2010, 2015). Hence, the projection emerging above the re-merger site of the embedded clause, βP in (130) cannot have a specifier above this landing site. Consequently, no Spec, βP c-commanding the both clauses is available if the embedded clause is a βP .

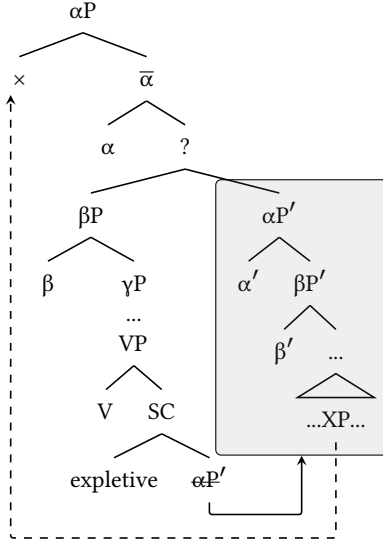
(130) **No Spec, βP above βP - $\beta P'$ re-merger**



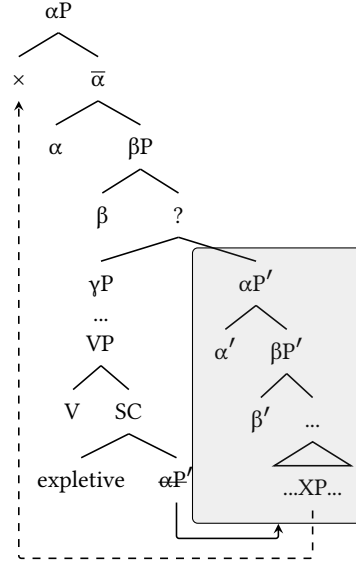
Although embedded clauses are capable of re-merging with non-equal-sized matrix projections, these asymmetric configurations do not make subextraction from embedded clauses possible. Firstly, MAIF in (125) rules out subextraction from embedded clauses that have been re-merged with a matrix projection that is smaller than their size. In (131), a larger $\alpha P'$ re-merges with a smaller βP . (*fseq* = $V < \gamma < \beta < \alpha$.) Here XP cannot be subextracted to Spec, αP even after its re-merger as XP would move along the ill-formed $*V' - \gamma' - \beta' - \alpha'$ sequence. Specifically, since two heads have the α label, this is not a well-formed *fseq*-fragment, i.e. MAIF is violated as the constant increase in *fseq*-indices does not hold. In (132), the embedded $\alpha P'$ merges with an even smaller matrix γP . In this case, subextraction from the embedded clause to matrix Spec, αP proceeds along the $*V' - \gamma' - \beta' - \alpha' - \beta - \alpha$ sequence, which includes a decrease in the *fseq*-indices and hence violates MAIF. To put it another way, since the $\alpha P'$ remains a maximal projection in the extended projection in

(131) and (132), no movement can leave it. Secondly, MAIF rules out subextraction in *smaller-to-larger* configurations, too. In (133), $\gamma P'$ re-merges with a larger βP . XP cannot move to the matrix Spec, αP because the sequence of heads crossed would be $*V'-\gamma'-\alpha$. This sequence is not a well-formed *fseq*-fragment because γ' and α are not adjacent in *fseq*; β is missing. Intuitively, the continuity of the extended projection is broken.

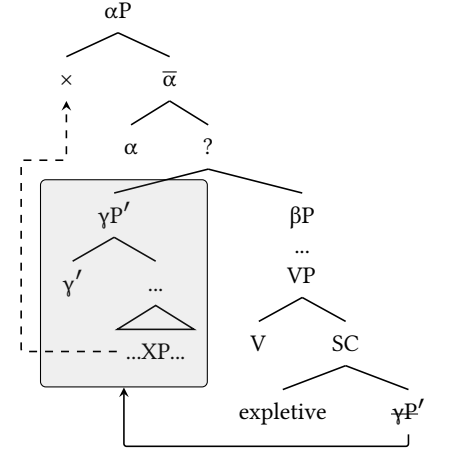
(131) *Larger-to-smaller*



(132) *Larger-to-much-smaller*



(133) *Smaller-to-larger*



In sum, Hungarian embedded clauses become transparent to subextraction after re-merging with an *equal-sized* matrix projection. The embedded clause's re-merger with an equal-sized matrix projection follows the more general idea of "unlocking by clause-external operations" (e.g. Rackowski and Richards 2005; Van Urk and Richards 2015). This contrasts with successive cyclic movement discussed in Section 5.2, which can be interpreted as "escaping by clause-internal operations".

In the next two sections, size- and position-dependent opacity in Hungarian will be shown to arise as a side-product of the properties of size-dependent clausal re-merger.

5.5 Deriving size-dependent opacity

Embedded clauses only become transparent to subextraction if they re-merge with an equal-sized matrix projection, and movement can land only land in specifiers c-commanding the landing site. It directly follows that smaller embedded clauses can re-merge with an equal-sized matrix projection lower whereas larger embedded clauses can re-merge with an equal-sized matrix projection higher. Lower matrix specifiers will only c-command the re-merger sites of smaller embedded clauses, so fewer types of subextraction will be possible from larger clauses and lower matrix clause positions will be harder to reach.

In formal terms, the point at which $\beta P'$ can re-merge with the matrix clause in a way that subextraction is MAIF-compliant from $\beta P'$ is when the matrix clause is also a βP . Notice that below this point, every matrix projection is a γP such that $\gamma < \beta$ in *fseq*; i.e. the re-merger of the embedded $\beta P'$ with these γP s would result in opaque *larger-to-smaller* configurations like (131) and (132). Once the $\beta P'$ re-merges with the equal-sized βP , every matrix specifier c-commanding this re-merger site is an instance Spec, αP such that $\beta < \alpha$ in *fseq*. Consequently, the generalization on size-dependent opacity (repeated in (134)) is derived if a language uses the re-merger of embedded clause to enable MAIF-compliant cross-clausal movement.

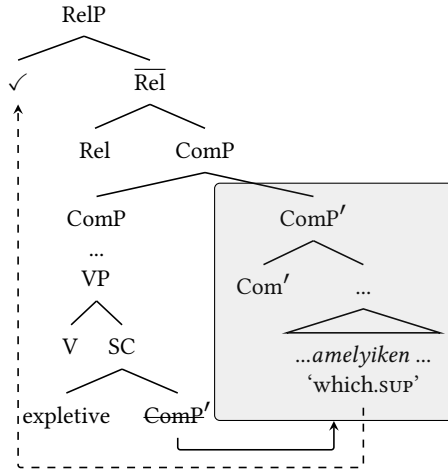
(134) **Size-dependent opacity/Strict Williams Cycle:** Movement to Spec, βP cannot proceed across a dominating αP , where $\alpha \geq \beta$ in the functional sequence.

The remaining part of this section shows some sample derivations from Hungarian. The clausal *fseq* established in Section 2 is the following:

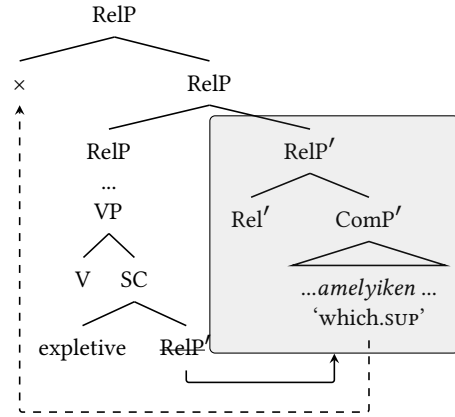
- (135) **Clausal fseq:** $V < sMod < T < Foc1 < Foc2 < Com < Rel$
 (in clauses with one focus, there is no Foc1-Foc2 distinction)

The generalization in (134) was drawn on the basis of the following movement types: Relative movement landing in Spec,RelP, focus movement landing in Spec,Foc2P and Spec,Foc1P, and particle movement landing in Spec,TP. Relative movement can proceed from ComPs and smaller clauses but not from RelPs. Since RelP is the highest projection of the extended projection, Spec,RelP c-commands embedded clauses that are re-merged with the matrix ComP or lower. The tree diagram in (136) demonstrates this with a ComP-ComP' re-merger. ComP' first moves along the well-formed V-sMod-T-Foc-Com sequence and re-merges with the equal-sized matrix ComP. Subsequently, the relative pronoun moves along the well-formed sequence V'-sMod'-T'-Foc'-Com'-Rel. That is, this derivation of cross-clausal relative movement is MAIF-compliant. Relative extraction from embedded clauses smaller than ComP will only differ from (136) in the height of the re-merger as the symmetric re-merger of a smaller clause happens lower (e.g. FocP' with FocP, TP' with TP, and sModP' with sModP). The impossibility of relative movement from RelPs in (137) follows from that no specifier projects above the re-merger site of embedded clauses i.e. no Spec,RelP projects above the re-merger site of RelP' just as we saw in (130).

- (136) **Relative subextraction from ComP**

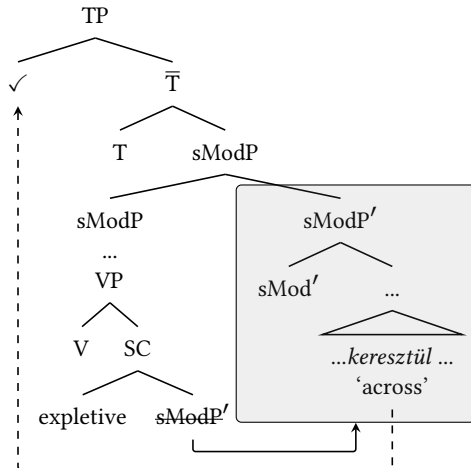


- (137) **No Spec,RelP projects upon the re-merger of RelP'**

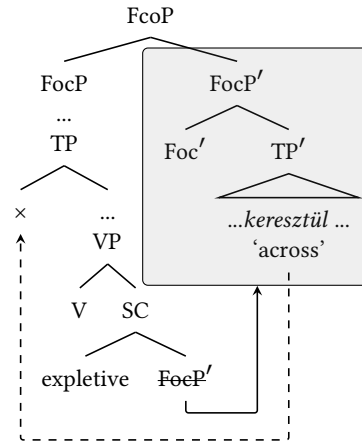


At the other extreme, the subextraction of particles is only possible from the smallest subjunctive clauses identified here as sModPs. In (138), sModP' re-merges with sModP in a way that the shared Spec,TP c-commands the embedded clause. (The movement of the clause happens within the matrix clause i.e. in a MAIF-compliant way.) After this, particles move along the well-formed V'-sMod'-T sequence.

- (138) **Particle subextraction from sModP**



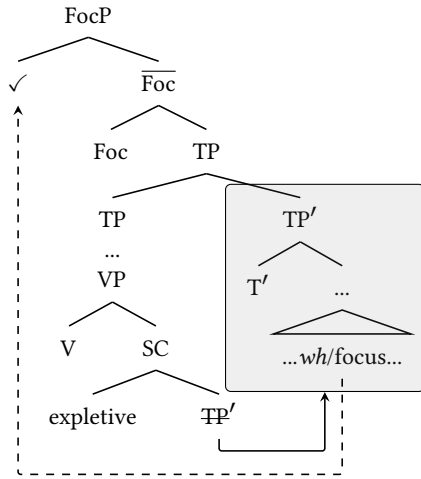
- (139) **No particle-subextraction if $\beta P' \geq FocP$**



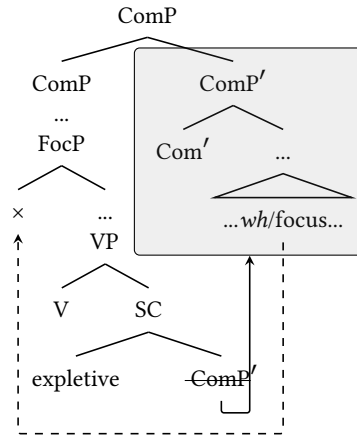
Larger clauses like FocPs and ComPs can also re-merge with the matrix sModP. However, since the emerging structures exemplify the *larger-to-smaller* configuration, such instances of re-merger do not facilitate the MAIF-compliant subextraction of particles. If embedded FocPs' or ComPs' do not leave the VP until the matrix clause reaches their sizes, particle subextraction is impossible because the matrix Spec,TP does not c-command the embedded clause. The impossibility of particle subextraction in such cases is demonstrated with an embedded FocP' in (139). Finally, in the same way, as no relative movement to Spec,RelP was possible from RelPs in (137), long particle movement to Spec,TP cannot proceed if another particle occurs in the embedded Spec,TP or the embedded clause is a tensed TP because no Spec,TP projects above the TP-TP' re-merger.

If the embedded clause is a TP or smaller, subextraction to matrix Spec,Foc is possible. For example, after TP' re-merges with matrix TP in (140), long focus movement proceeds along a MAIF-compliant sequence of heads (V'-sMod'-T'-Foc). The analysis is similar for smaller embedded clauses like sModPs except that these must be attached lower. On the other hand, focus extraction is impossible from ComPs and RelPs. In the system presented here, this is predicted as in order to be open for MAIF-compliant movement, embedded ComPs and RelPs need to re-merge above FocP as illustrated with a ComP in (141). Therefore, no Spec,FocP c-commands both clauses. Again, if an embedded ComP' re-merges at matrix TP or lower, we are dealing with a *larger-to-smaller* configuration, from which subextraction violates MAIF.

(140) **Focus/wh-subextraction from TP**



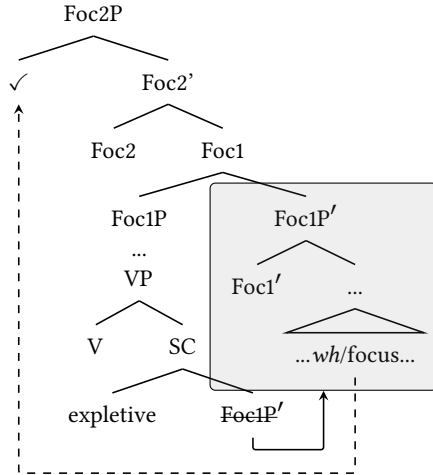
(141) **No focus/wh-subextraction from ComPs**



In biclusal structures in which a focus/wh-constituent moves across another, Foc1P' re-merges with Foc1P in a way that the higher Spec,Foc2P c-commands Foc1P' as in (142). In this way, XPs subextracted after the re-merger of Foc1P' will move along the V'-sMod'-T'-Foc1'-Foc2 sequence, which is MAIF-compliant. Notice that just as the RelP-RelP' re-merger did not allow relative extraction in (137) because no Spec,RelP projected above the re-merger site, the Foc2P-Foc2P' re-merger does not allow focus extraction as no Spec,Foc1P or Spec,Foc2P projects above the re-merger site.³²

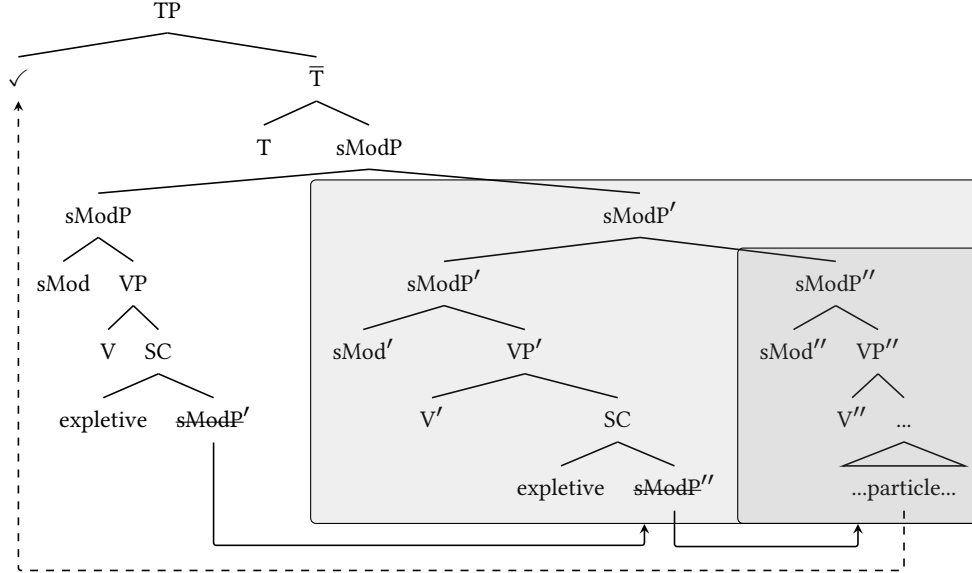
³²As I pointed out in Footnote 4, the number of focus projections is not limited to two in Hungarian, i.e. recursion can in principle bring about a Foc3P, a Foc4P etc. In order to deal with this problem, we could introduce multiple tracks of *fseq*-indices. The main track would track the category labels sMod, T, Foc, Com, Rel etc.; the secondary track would track sub-indices e.g. Foc1, Foc2, Foc3. In this way, a well-formed *fseq* would tolerate adjacent heads with the same main index if the sub-index increases. In this way, sMod-T-Foc-Com; sMod-T-Foc1-Foc2-Com; sMod-T-Foc1-Foc2-Foc3-Com would be well-formed sequences but *sMod-T-Foc1-Foc1-Com, *sMod-T-Foc3-Foc2-Com, and *sMod-T-Foc1-Foc3-Com would be bad indices. Since I do not discuss clauses with more than two foci in the main text, I set this problem aside.

(142) **Focus/wh-extraction from FocP**



As demonstrated in Section 2.3, the Williams Cycle holds to multiple embeddings as well. For instance, example (45) indicated that particle movement is possible across multiple clause boundaries unless one of these clauses turns out to be larger than sModP. In the derivation in (143), two clauses move: First the deepest sModP'' re-merges with the intermediate sModP', then the phrase emerging immediately above these re-merges with the matrix sModP. Neither of these movement steps crosses any extended-projection boundary, that is, they are MAIF-compliant. Henceforth, the particle in the deepest sModP'' can move as the sequence of crossed heads is V''-sMod''-T, that is, the derivation is MAIF-compliant. Here the rationale is that after the two re-mergers the matrix TP serves as a continuation not only for the extended projection of the intermediate clause but also for the extended projection of the deepest embedded clause.

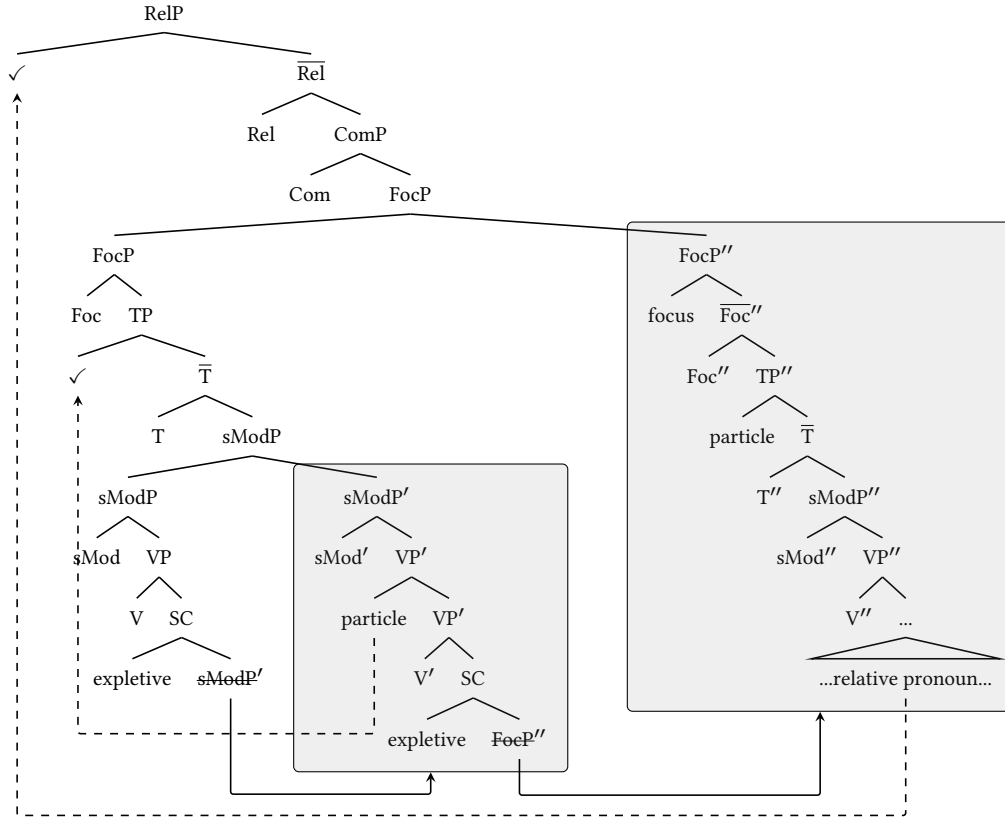
(143) **Particle movement across multiple clauses**



In (47), the intermediate clause must be an sModP' because a particle gets subextracted from it, whereas the deepest clause has a focus, that is, it must be a FocP''. In other words, the deepest clause is larger than the intermediate clause. The derivation of the subextraction of the particle and the relative pronoun is shown in (144): The intermediate sModP' first re-merges with the matrix sModP, which is a MAIF-compliant movement within an extended projection. Subsequently, the movement of the particle to matrix Spec,TP is MAIF-compliant in the same way as (138): the sequence of heads crossed is V'-sMod'-T. After this, the deepest FocP'' re-merges with the matrix FocP. The movement of FocP'' is also MAIF-complaint because the sequence of heads crossed is V'-sMod'-T-Foc. After this, the subextraction of the

relative pronoun from the deepest clause proceeds along the well-formed V'' - $sMod''$ - T'' - Foc'' - Com - Rel sequence. In sum, all movement steps in (144) are MAIF-compliant. ³³

(144) **The deepest clause re-merges with the matrix clause**



This paper does not provide an analysis for violations of (116) not clearly derivable from successive cyclicity. Furthermore, unbounded successive cyclicity requires movement steps across αP to another $Spec, \alpha P$, e.g. from CP to $Spec, CP$. As discussed in Section 2.5, such successive cyclic movement steps are excluded under the Strict Williams Cycle in (116) but not under the original Williams Cycle (Williams 2003, 2013) in (145). As demonstrated with Mongolian in (117), movement to the highest specifier of some domain makes any landing site available in the next one. Thus, my paper cannot derive the ban on hyperraising from CP to $Spec, TP$ in English if English does indeed exhibit successive cyclic movement from $Spec, CP$ to $Spec, CP$, as widely assumed. Another language exhibiting reflexes of successive cyclicity as well as Williams Cycle effects is German (Keine 2016, 2020).

(145) **Williams Cycle:** Movement to $Spec, \beta P$ cannot proceed from $Spec, \alpha P$ or across αP , where $\alpha > \beta$ in the functional sequence. (based on Poole 2023; Williams 2003)

³³ Throughout this paper, it has been assumed that heads between V and Rel belong to a single *fseq*. The analysis keeps working if we break this *fseq* into two in a way that v is the highest head in the lower *fseq*, that is, vP is a locality domain on its own. In (i), the embedded $\beta P'$ moves first to the matrix $Spec, vP$. Subsequently, since the movement to $Spec, vP$ does not bring about criterial freezing, the embedded $\beta P'$ re-merges to the left of matrix βP . The subextraction of XP to matrix $Spec, \alpha P$ proceeds through the embedded $Spec, vP'$. After leaving $Spec, vP'$, the sequence of heads XP crosses before landing in $Spec, \alpha P$ is $\gamma' - \beta' - \alpha$, that is, the movement is MAIF-compliant.

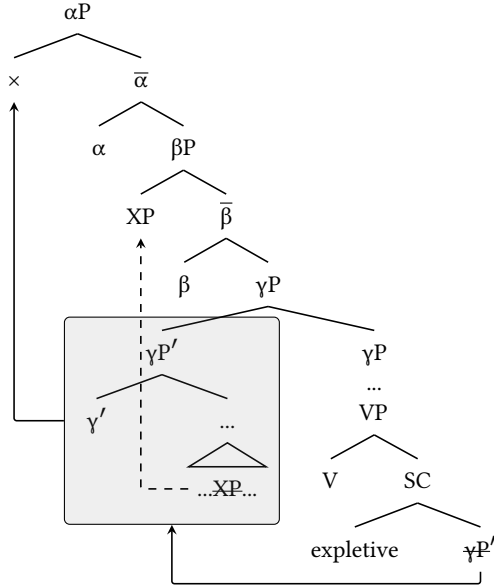
(i) $[\alpha P \checkmark \alpha [\beta P [\beta P' \beta' [\gamma P' \gamma' \dots [\gamma P' XP v' [\gamma P' V' \dots XP \dots]]]] [\beta P \beta [\gamma P \gamma \dots [\gamma P \beta P' v [\gamma P V \dots \beta P' \dots]]]]]$

Thereby, even in the presence of intermediate stops in $Spec, vPs$, the analysis presented so far predicts Williams Cycle effects within the clausal *fseq*. This analysis of Hungarian is compatible with successive cyclicity exclusively through $Spec, vPs$ as it has been suggested by Den Dikken (2009). In the remainder of this paper, I ignore vP -phases. The analysis applicable to any other non-clausal functional sequence, for instance, Section 5.2 mentions that Hungarian DPs can be escaped successive cyclically. Lipták (2009) mentions that relative clauses can escape DPs in left-periphery positions in a way that the relative clauses end up in the right-periphery. Under the analysis presented here, the relative clause can re-merge with the matrix clause in a MAIF-compliant way via the edge of the DP.

5.6 Deriving position-dependent opacity

Section 5.4 introduced the idea that in the absence of successive cyclicity, Hungarian $\gamma P'$ clauses become transparent to the movement of XP to some Spec, βP (such that $\gamma < \beta$ in *fseq*) only if $\gamma P'$ re-merges with the equal-sized matrix γP projection as in (146).

(146) **The re-merged clause cannot move any further**



As shown in (146), the node emerging from $\gamma P'$ and γP is a γP as both of the lower γP s project. I propose that the higher γP not only inherits the shared category label of the two γP s (as in Chomsky 2013), but also the union of the feature matrices of the two γP s. In other words, $\gamma P'$ and γP are no longer maximal projections once another γP emerges above them. The ban on moving non-minimal and non-maximal (bar-level) projections in (147) is a standard assumption in the syntactic literature. Apart from minimal projections, the computation and the interfaces in the *Minimalist Program* can only see maximal projections, which do not project further (Chomsky 1995).³⁴

(147) Projections that are neither minimal nor maximal cannot undergo movement.

More generally, once MAIF only permits subextraction from embedded clauses that have re-merged with equal-sized matrix projections, gaining transparency comes with the loss of mobility: once re-merged $\gamma P'$ cannot subsequently move to Spec, αP in (146) because $\gamma P'$ is dominated by a maximal γP . As a consequence, no clause can move to a position c-commanding material extracted from it. In other words, the generalization on position-dependent opacity (repeated in (148)) is derived.

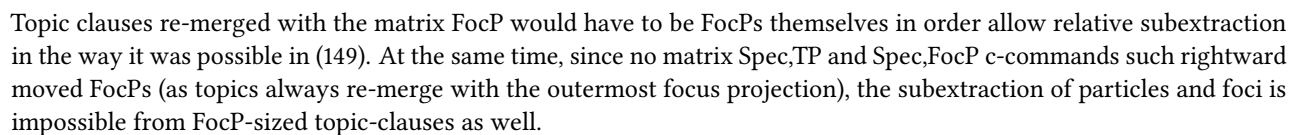
(148) **Position-dependent opacity:** αP or the small clause containing αP cannot move to a position immediately dominated by a βP , if some XP from αP moved to Spec, γP , and $\alpha, \beta, \gamma \in fseq F$, and $\beta \geq \gamma$.

As demonstrated in Section 5.2, remnant movement is not ruled out in general by the MAIF-based account: If a certain extended projection can be escaped via successive cyclic movement, no re-merger is needed, so nothing stops the remnant movement of the extended projection.

This analysis can be applied to the Hungarian data presented in Section 3.3. The diagram in (149) represents a sentence in which an embedded clause $Comp'$ is moved to the front and gets a topic-interpretation. After moving along the well-formed V-sMod-T-Foc-Com sequence, $Comp'$ re-merges with an equal-sized $Comp$. Since the embedded clause is moved to the left, it is sandwiched between the relative pronouns and foci in the linear order. Thus, this re-merger site asymmetrically c-commands Spec,FocP and Spec,TP, as the leftward movement of clauses cannot land within a specifier dominated by some FocP (discussed in Section 3.2.2 Kenesei 1994). Consequently, focus and particle subextraction are

³⁴Possible ways to explain this restriction are *Relativized Minimality* (Rizzi 1990) and the *Minimal Link Condition* (Chomsky 1995): Since a c-commanding probe's search must be minimal, the mother γP is a closer goal than its $\gamma P'$ daughter as γP has all the features that $\gamma P'$ has.

(149) **Subextraction from a clause must land higher than its re-merger site**

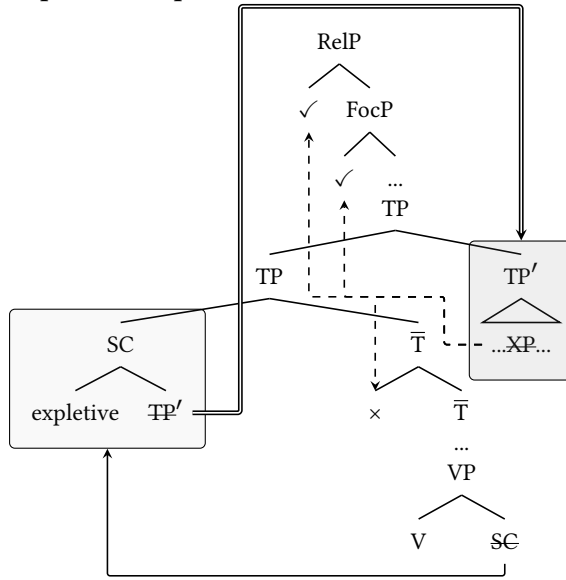


The analysis of sentences where the expletive of the embedded clause shows up in the matrix left-periphery involves the movement of the entire small clause. In Section 3, I introduced two constraints regarding expletive positions, these are repeated in (150) and (151).

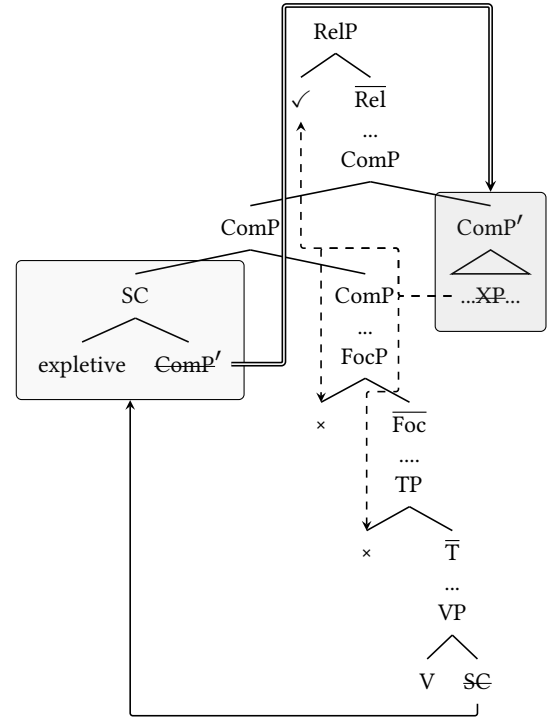
- In the context of clausal expletives, the generalization on position-dependent opacity in (148) can be reformulated in the following way: Subextraction from an embedded clause cannot land lower in the left-periphery than its small clause associated with the clause containing the expletive. The general idea behind the analysis is that under (150), the embedded clause cannot leave the small clause until the small clause reaches its final position, i.e. the re-merger site of the clause always c-commands the expletive. Since the landing site of subextraction needs to c-command the embedded clause and the re-merger site needs to c-command the expletive, (148) falls out from the transitivity of the c-command relation, i.e. subextraction can only land in a position c-commanding the expletive.

Section 3 introduced examples with overt expletives contained in Spec,TP, Spec,FocIP, Spec,Foc2P as well as in positions sister to CompPs and FocPs. In (152), the entire small clause undergoes movement to Spec,TP along the well-formed V-sMod-T sequence. Since no finite clause is tolerated within Spec,TP, the embedded TP' immediately re-merges with the matrix TP by (151). Since the expletive does not count as a head, this re-merger is MAIF-compliant. (Here I use double line for subextraction from the small clause.) In this way, subextraction to the matrix Spec,FocP and Spec,RelP proceeds along the well-formed V'-sMod'-T'-Foc and V'-sMod'-T'-Foc-Com-Rel sequences, respectively. However, since no matrix Spec,TP c-commands the re-merger/adjunction site of TP', particle subextraction from the embedded clause is impossible. Since the embedded clause must c-command the expletive throughout the derivation by (150), the embedded clause cannot re-merge in a lower position than is asymmetrically c-commanded by the final Spec,TP position of the small clause (e.g. no sModP-sModP' re-merger is possible).³⁵ Finally, the embedded clause can be larger than TP and re-merge higher. For example, if the embedded clause is a CompP', it can undergo movement from within Spec,TP along the Foc-Com sequence and re-merge with the matrix CompP. In this case, focus subextraction becomes impossible just as it was in (149).

(152) **Expletive in Spec,TP**



(153) **Contained within a CompP-sister**



If the small clause lands higher in the left-periphery, the embedded clause needs to re-merge higher. In (153), the small clause re-merged with the matrix CompP as a topic. From here, the first matrix projection with which the embedded clause can re-merge is the matrix CompP. If it does so, subextraction to Spec,RelP only crosses Rel head, i.e. the movement is MAIF-compliant. On the other hand, since the embedded clause needs to c-command the expletive at every point in the derivation, even if the embedded clause was smaller (e.g. sModP), its re-merger with a lower equal-sized matrix projection would freeze the embedded clause in that lower position, which would rule out the movement of the small clause remnant to a CompP-sister position. Since the re-merger site cannot be c-commanded by the matrix Spec,TP or Spec,FocP, the subextraction of particles and foci is impossible from the embedded clause in (153).³⁶

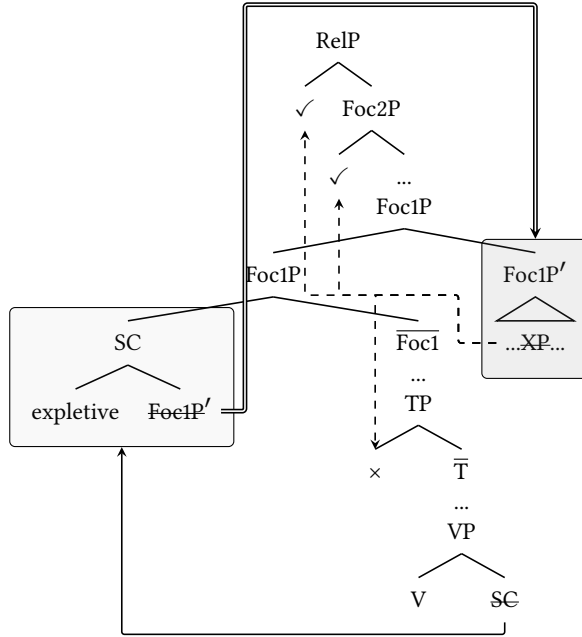
³⁵Notice that the embedded clause's re-merger precludes the further movement of the embedded clause (see (147) in the previous subsection). Therefore, even if (150) did not have to hold at every derivational step but was a condition on complete tree structures, the remnant movement of the small clause to a higher position than the re-merger site of the embedded clause could not be followed by the movement of the re-merged clause into a c-commanding position, i.e. (150) would be violated.

³⁶In structures like (71), the embedded clause can remain pre-verbal within the CompP/FocP-sister small clauses. Assuming the expletive is not a head, the extraction possibilities are the same in such cases as for clauses re-merged with CompP/FocP in (149) because the set of c-commanding heads is the predicted to be the same. This is borne out: Irrespectively of the presence of the expletive before the FocP/CompP-sister clause (i.e. whether the small clause or just the embedded clause moves), particle subextraction is impossible in (88), focus subextraction is impossible in (89), but relative subextraction is possible in (90). In theory, the embedded clause could undergo further movement from its pre-verbal position, but the testing

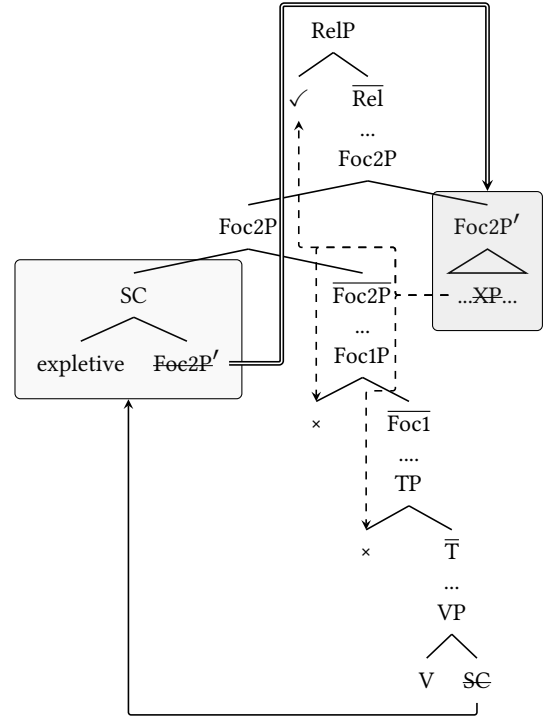
As it was indicated by the clausal expletives in Section 3, small clauses can be focused i.e. they can move to a Spec,Foc1P or Spec,Foc2P specifier. In (154), the small clause moves to Spec,Foc1P, which is followed by the re-merger of the embedded Foc1P' with the matrix Foc1P (by (151)). In (155), the small clause moves to Spec,Foc2P, which is followed by the re-merger of the embedded Foc2P' with the matrix Foc2P (again, by (151)). As far as particle and relative subextraction is concerned, the transparency of the embedded clauses is the same in (154) and (155): in both cases, the equal-sized re-merger asymmetrically c-commands the matrix Spec,TP and is c-commanded by the matrix Spec,RelP. Thus, in both cases, MAIF-complaint particle subextraction is impossible. Again, the embedded clause cannot be smaller and re-merge lower as it must c-command the expletive. At the same time, relative subextraction can proceed along the well-formed sequence V'-sMod'-Foc1'-Foc2(')-Com-Rel; the only difference is whether the movement crosses the matrix or the embedded Foc2 head. (To put it in another way: As far as particle and relative subextraction are concerned, we could replace Foc1P and Foc2P with a single FocP projection and a FocP-FocP' re-merger.)

The structures in (154) and (155) differ with respect to the possibility of focus sub-subextraction. In (154), Foc1P' re-merges with the matrix Foc1P in a way that it is still c-commanded by the matrix Spec,Foc2P. Consequently, MAIF-complaint focus subextraction can proceed along the V'-sMod'-T'-Foc1'-Foc2 sequence. In (155), no matrix Spec,FocP c-commands the Foc2P-Foc2P' merger site, so no focus extraction is possible. FocP-sister clauses have the same hierarchical position as the minimal landing site of clauses moved out from Spec,Foc2P; this is why the extraction-possibilities are the same in the two cases.

(154) **Expletive in Spec,Foc(1)P**



(155) **Expletive in Spec,Foc(2)P**



In sum, the idea behind the analysis here is that since the embedded clause needs to c-command the expletive associated with it at each step of the derivation, the embedded clause can only re-merge with the matrix clause after the small clause containing the embedded clause has reached its final position. Consequently, MAIF-complaint movement can only land in a position c-commanding the clausal expletive. The intuition connecting this section to the previous one is that once the embedded clause becomes transparent via re-merger with an equal-sized matrix projection, the embedded clause or the small clause containing it cannot undergo further movement.

possibilities are limited as there is just one specifier (Spec,RelP), and it c-commands every clause-position in the left-periphery.

6 Conclusion

In this paper, I have first shown that Hungarian exhibits *size-dependent opacity*: Clauses with larger left-peripheries were opaque to more movement types, and the lowest specifiers of the matrix left-periphery can be reached from only from syntactically smallest embedded clauses. In addition, Hungarian exhibits *position-dependent opacity*: Neither clauses nor the small clauses containing them can be higher in the left-periphery than material extracted from these clauses. Since position-dependent opacity cannot be reduced to size-dependent opacity, previous accounts of size-dependent opacity do not provide a unified account.

The key ingredients for my analysis of the Hungarian size- and position-dependent opacity were the following: (i) syntactic movement is constrained by MAIF, i.e. movement must be internal to extended projections³⁷; (ii) embedded clauses are immobile after they re-merge with an equal sized matrix projection; (iii) the embedded clause must c-command its expletive at every derivational step; (iv) there is no specifier above the re-merger of clauses (optional operations follow obligatory operations); and (v) there is no successive cyclic movement in Hungarian clause edges.

Due to space reasons, some questions could not be addressed in the current discussion. Firstly, assumption (v) says that successive cyclicity is in a complementary distribution with size- and position-dependent opacity: The Hungarian clausal domain had size- and position-dependent opacity because no movement could escape them successive cyclically. In contrast, a language exhibiting size- or position-dependent opacity as well as reflexes of successive cyclicity in a single extended projection cannot be derived by the analysis presented here. For example, Keine (2016) discussed that German exhibits reflexes of successive cyclicity as well as Williams Cycle effects. Secondly, my account focused on the derivation of the strict Williams Cycle, which ruled out subextraction from β Ps to Spec, β P. It remains an open question how the original Williams Cycle (Williams 1974, 2003, 2013), which permits such movement steps, can be derived by a MAIF-based account. Finally, my analysis in its current form only deals with the Williams Cycle and selective opacity in the context of syntactic movement. Hence, my analysis cannot explain known cases of Williams Cycle effects in case (e.g. Poole 2023) and Agree (e.g. Keine 2016; Egressy to appear). It is possible that the Williams Cycle data involving Case and Agree involve another source than my movement-oriented data.

The relative novelty of the MAIF-based theory lies in that it provides a unification for the standard notions of locality (subjacency, barriers, phases, successive cyclicity etc.) and the various kinds of selective opacity (but also see Müller 2014a,b). The connection between these is not trivial (Keine 2019): While the existing theories of locality (e.g. the Phase Theory) are generally concerned with the longest grammatical movement dependency, the constraints of *size-* and *position-dependent* opacity are restrictions on the relative heights of launching and landing sites across clauses.

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³⁷A MAIF-free alternative to my proposal is to stipulate that instead of moving, complement clauses are base-generated as sisters of a matrix clause position of their size, and from this position they bind their expletives. In this way, size-dependent opacity would be derived because the matrix specifiers asymmetrically c-commanded by the base-generation site of an embedded clause would be unavailable from this clause. Position-dependent opacity would be derived from the symmetry of the matrix-embedded re-merger as in Section 5.6: Embedded clauses would be unable to leave their base-generated positions because they are dominated by a featurally identical node. Nevertheless, this alternative has its own problems. Firstly, embedded clauses reconstruct to (low) VP-internal position for Principle C as demonstrated in Section 3.1. Moreover, the verbs' syntactic selection for various clause-types (cf. Section 2.4) would be difficult to encode in the absence of VP-internal clauses. The question whether embedded clauses are base generated low or clausal complementation involves the high base-generation of clauses and the binding of expletives is also discussed by É. Kiss (2023), who analyses complementation in Modern Hungarian as binding but that in earlier stages as low base-generation and movement. At the same time, even if complement clauses were base-generated VP-internally after a certain point in É. Kiss's (2023) analysis, the movement of clauses keeps rendering symmetric CP-CP structures up until today (also see Lipták 2009). My proposal differs from this in that embedded clauses come in various sizes and they can re-merge with various matrix projections, hence, any embedded clause can re-merge with a matrix projection of its own size.

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